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While chatting about number theory problems, I'm often asked about the Collatz Conjecture. Many of my non-math friends asked me why its so hard, and I couldn't answer them. My mathematical colleagues couldn't answer either besides saying Terry Tao said it was hard. This paper summarizes a little project I've done in trying to understand why the Collatz Conjecture is still an unsolved problem. In particular, I would like to find a reformulation of the Collatz Conjecture that shows that the simple looking statement is not as innocent as it seems.

Part of the goal was to just explore as far as possible and see what I will find. This culminated in finding a probability distributions that can be put on a number that is invariant to Collatz function up to a small degree, and using this probability distribution we can look for the expected values in two important functionals that shall be outlined in this paper.

The "small degree" mention above turns into the key issue making the Collatz Conjecture so hard to solve: I showed that if the end of the chain representation of a number (outlined in this paper) has certain conjectured probabilistic values, then we can provably show the probability of a number *not* reaching 1 in finite iterations is 0%. The fact of the mater is, it is rather hard to show this final result, and I outline my current best attempt at it in this paper.

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Main Paper

Abstract

Inspiration All theorems in this paper have been deduced by looking at a ton of empirical data and then proving the patterns that were found. This is especially true for the results about the difference in length which where easy to verify empirically but were harder to prove.

Let $n \in \mathbb{N}$. Consider the following arithmetic progression:

$$f(n) = \begin{cases} \frac{n}{2} & n \text{ is even} \\ 3n+1 & n \text{ is odd} \end{cases} \tag{1.1}$$

Then the Collatz conjecture claims that there exists a k such that $f^{\circ k}(n) = 1$ for every n. With the goal of understanding this result better, I shall present a re-framing of the Collatz Conjecture in the following steps that lead to a new conjecture that demonstrates more precisely the difficulty:

1. To gain more information out of each number, I shall look at the array of the difference in exponents when a number is represented as a sum of 2-powers; these shall be called the *2-chain representation* of a natural number, or just *chain representation* for short (see definition 1). This is not the only way that we can gain more information from a natural number, in fact for the proof that will be presented in this paper, there are many formulations that can be used, however this representation was chosen specifically as it will be illustrative on where the bottleneck in solving the Collatz Conjecture arises

   After finding the chain representation, we shall defined *Collatz Arithmetic's*, which will be the arithmetical rules that will allow us to algorithmically find the next chain representation of a number without needing to reference the Collatz function or natural arithmetic, essentially translating the Collatz Conjecture into a problem about chains and Collatz Arithmetic.

2. We shall next define some functionals on chains that shall extract useful information. Two natural functional we may define are the *length* functional $m(-)$, and *size* functional $\sum(-)$, where the first measure the number of positions of a chain while the other sums over all the positions. We shall prove that the size functional shall have the following property: if N_k is the chain representation of $f^{\circ k}(n_0)$, and $F(N_k)$ is the chain representation of the next odd Collatz number, then:

   $$\sum F(N_k) = \sum N_k + \{-1, 0, 1\} - 2\ell$$

   where $\ell \in \mathbb{N}$ will be the number of 2's at the beginning of the chain. In other words, a chain will increase in size by at most 1, and has many opportunities to shrink. We shall show that the size only depends on the *first digit* and *last block* of the chain, where we shall define what is a block within this paper; it shall usually be the last few digits of a chain.

   Thus, the Collatz Conjecture can be re-framed as asking to show that the size of a chain will always reach 0.

3. With the goal of showing that the size of any the chain shrinks to 0 after applying a finite iteration of the Collatz function, we need to find a way to determine the value of the first digit and last few digits of a chain, as well as the expected change in length of a chain. This is where we shall introduce a probabilistic distribution on the digits of a chain. To understand where this distribution comes from, consider first any natural number $n \in \mathbb{N}_{>0}$. Representing it as a chain C_0, we can show that the probability that the ith digit $C_{0,i}$ of the chain is equal to is 1, 2, or ≥ 3 is:

   $$P(C_{0,i} = 1) = 0.5 \qquad P(C_{0,i} = 2) = 0.25 \qquad P(C_{0,i} \geq 3) = 0.25 \tag{1.2}$$

   We shall show that these probabilities are stable after applying the Collatz function *away from the tail.*

   Thus, the Collatz Conjecture can be re-framed to requiring the first digit and last block to have the right probabilities so that the size of any chain must eventually reach 0.

4. To show stability away from the tail, we show the result on infinitely long chains, which are a subset of 2-adic numbers that behave well in the aforementioned probabilistic model. These shall have invariant probability under Collatz function.

   In the process of proving this invariance we shall notice that there are two "types" of probabilities to keep track of. There is the *local probability* of a chain, which is the probability that a given digit in a chain C will be altered or consumed (which will be a term defined later). There is then the *global probability* which is the probability of a new digit appearing or an old digit diapering somewhere in the chain (not necessarily at any particular position). For an infinitely long chain, these two shall interact well since the global behaviour is "uniform", but for finite chains, the tail of the chain behaves differently requiring a different approach.

   Nevertheless, the case of the infinite chains will allow us to conclude that the expected value of the difference in sum and difference in length functionals on infinite chains are *negative.*

5. Finally, I shall give my best guesses on how these methods can be applied to finite chains. These are (naturally) incomplete; completing them would mean the Collatz conjecture would have been solved. I shall give my own conjecture on what is needed for the Collatz conjecture to be solved which shall be my guess on what the probability of the last digit should converge to.

1.1 Chains and Collatz Arithmetic's

First, we must represent each natural number t as a *chain*. This chain will give way to *Collatz arithmetic's*, which will be important in computing various aspects of the probability problem. We may start with an easy simplification: if n is even, divide it until it's odd. Then if $t = 1$, we're done, so assume it's not the case. Represent n in the following way:

$$t = 1 + 2^{n_1}(1 + 2^{n_2}(1 + \cdots))$$

For example:

$$77 = 1 + 2^2(1 + 2(1 + 2^3))) = 1 + 2^2 + 2^3 + 2^6$$

Simplifying notation, we may write this as $\sim\ 2\ \sim\ 1\ \sim\ 3$ (representing the difference in power's of 2). This can be done for nay natural number. This will be called the *chain representation*

Definition 1: Chain Representation

Let $t \in \mathbb{N}_{>0}$. Then the *chain representation* of n is the string

$$t_0\ \sim\ t_1\ \sim\ t_2\ \sim\ \cdots\ \sim\ t_k$$

where each t_i is the difference in powers in the binary representation. The string will be called a *chain*, and each number in the chain is called an *element* or *component* of the chain. The chain $\sim\ 0$ shall often be represented as $\sim$. If n is odd, then it will be represented as:

$$\sim\ t_1\ \sim\ t_2\ \sim\ \cdots\ \sim\ t_k$$

In the following, we will examine the behavior of a chain as it passes through f (see equation 1.1). As we shall see, the current representation only depends on what happens when n is odd, so define the function F to take in an odd number and return the next odd progression, for example if $n = 13$, then $F(13) = 5$. The collection of all progressions shall sometimes be of interest:

Definition 2: Collatz Sequence

Let t be a number which without loss of generality is odd. Then if N_0 is the chain representation of t, the sequence

$$(N_0, F(N_0), F(F(N_0)), ...)$$

is called the *Collatz sequence* with respect to t (or for t). Often, the shorthand $N_k = F^{\circ k}(N_0)$ will be used, and the sequence may be represented as $(N_i)_i$ if t is clear.

Note that the Collatz sequence is always infinite, however if the Collatz Conjecture is true this sequence shall at some point loop between

$$1 \quad \Rightarrow \quad 4 \quad \Rightarrow \quad 2 \quad \Rightarrow \quad 1 \quad \Rightarrow \quad \cdots$$

which in terms of chains would be represented $F(\sim) = \sim$. Hence given the Collatz Conjecture is true any sequence will eventually be:

$$(\cdots, \sim, \sim, \sim, \cdots)$$

1.1.1 Collatz Arithmetic

What we shall next endeavor is to fully classify the behavior of chains when passing through F once. There is a computational advantage in doing this: computing the next number in the Collatz sequence for very large numbers becomes intractable for very large numbers. For example the next Collatz number represented by $\sim 3 \sim 10'000 \sim 1$ is larger than 2^{10000}, however by doing Collatz arithmetic it takes a single computer cycle to deduce the next number is represented by $\sim 1 \sim 1 \sim 9'999 \sim 3$.

Collatz Arithmetic shall be completely dependent on whether a component has value 1, 2, or $n \geq 3$. Starting with the case where a chain only consists of 1, after passing through F:

$$\sim 1 \ \sim \ 1 \ \sim \cdots \sim \ 1 \ \sim \ 1 \ \Rightarrow \ \color{red}{\sim \ 1} \sim 1 \ \sim \cdots \sim \ 1 \ \sim \ 1 \color{blue}{\sim \ 2} \ \equiv \sim \ 1 \sim \cdots \ \sim \ 1 \ \sim \ 1 \ \sim \ 2$$

We see that the first one gets "consumed", while a new power of two gets added at the end of the chain:

Lemma 1: Chain of 1

Let t be the odd number represented by a chain of ones: Let $t_i = 1$ for each i. Then:

$$F(\sim \ t_1 \ \sim \cdots \ \sim \ t_k) = \sim \ t_2 \ \sim \ \cdots \ \sim \ t_k \ \sim \ 2$$

Proof **:**
In binary representation, this number is of the form:

$$1 + 2 + 2^2 + \cdots + 2^{t_k}$$

Tripling and adding one, we get:

$$1 + (1 + 2 + 2^2 + \cdots + 2^{t_k}) + (2 + 2^2 + 2^3 + \cdots + 2^{t_k+1}) = 2 + \cdots + 2^{t_k} + (2^{t_k+1} + 2^{t_k+1})$$

Dividing by 2 and transforming to the chain representation, we get:

$$\underbrace{\sim \ 1 \ \sim \cdots \sim \ 1}_{(t_k - 1) \text{ times}} \ \sim \ 2$$

as we sought to show.

If we had a single $\sim \ 1$, then we would get $F(\sim \ 1) = \sim \ 2$. Notice the first one always disappears. For reasons that shall be clear soon, we shall say that this one was *consumed* to start the *active propagation*. These terms shall be made rigorous soon, the vocabulary is introduced now to ease the reader into it. What if we have an arbitrary chain of only 2's? This is our ideal result:

Lemma 2: Chains Of 2

Let t be the number represented by a chain of two's: let $t_i = 2$ for each i. Then:

$$F(\sim \ t_1 \ \sim \cdots \ \sim \ n_k) = \sim$$

***Proof* :**
In binary representation, this number is of the form:

$$1 + 2^2 + 2^4 + \cdots + 2^{2t_k}$$

Tripling and adding one, we get:

$$1 + (1 + 2^2 + 2^4 + \cdots + 2^{2t_k}) + (2 + 2^3 + 2^5 + \cdots + 2^{2t_k+1}) = 2^{2t_k+2}$$

But this is a power of two, hence dividing away we get $F(t) = 1$, as we sought to show.

More generally, for any chain that starts with sub-chain of 2's, the then 2's will "disappear"

Lemma 3: Disappearing 2's

Let t be the number represented by the chain $\sim\ 2\ \sim \cdots \sim\ 2\ \sim\ C$ where C represents some arbitrary other chain. Then:

$$F(\sim\ 2\ \sim \cdots \sim\ 2\ \sim\ C) = F(C)$$

***Proof* :**
In binary representation, this number is of the form:

$$1 + 2^2 + 2^4 + \cdots + 2^{2t_k} + 2^{2n_k+1/n} + \cdots$$

where the next digit is either 1 more the power of the previous one, or a power $n \geq 3$ more than the previous one. Tripling and adding one, we get:

$$\begin{gathered} 1 + (1 + 2^2 + 2^4 + \cdots + 2^{2t_k} + 2^{2n_k+1/n} + \cdots) \\ +(2 + 2^3 + 2^5 + \cdots + 2^{2t_k+2} + 2^{2t_k+2/(n+1)} + \cdots) \\ = 2^{2t_k+2} + \cdots \end{gathered}$$

The beginning of the chain a power of two, hence dividing away we get $F(C)$, as we sought to show.

This is rather special. The case of the first 2's of a chain disappearing will be called the *pre-active propagation.*

Let us now take a moment to analyze the case of a chain with an arbitrary number of $n \geq 3$, the behavior each n may differ slightly, and so we first tackle the case where we only have a single n:

Lemma 4: Single n

Let t be the number represented by the chain $\sim\ n$ for some $n \geq 3$. Then:

$$F(\sim\ n) = \sim\ (n-2)\ \sim\ 1$$

***Proof* :**
In binary representation, this number is of the form:

$$1 + 2^n$$

Tripling and adding one, we get:

$$1 + (1 + 2^n) + (2 + 2^{n+1}) = 2^2 + 2^n + 2^{n+1}$$

Dividing by 4, we get $1 + 2^{n-2} + 2^{n-1}$, which in chain representation is:

$$\sim\ (n-2)\ \sim\ 1$$

as we sought to show

This process adds a new number in our chain, namely a 1. We shall say that the n has *split*. Note that the splitting happened to the *right*; this shall always be the case. Let us next consider a chain where each element is ≥ 3:

Lemma 5: Chain of n's

Let t be the number represented by a chain

$$\sim\ n_1\ \sim\ n_2\ \sim \cdots \sim\ n_k$$

where $n_i \geq 3$ for each i. Then:

$$F(\sim n_1 \sim n_2\ \sim \cdots \sim n_k) = \sim (n_1{-}2)\ \sim\ 1\ \sim (n_2{-}1)\ \sim\ 1 \sim\ (n_3{-}1)\ \sim \cdots \sim\ (n_k{-}1)\ \sim\ 1$$

***Proof* :**
In binary representation, this number is of the form:

$$1 + 2^{n_1} + 2^{n_1+n_2} + \cdots$$

Tripling and adding one, we get:

$$1{+}(1{+}2^n_1{+}2^{n_1+n_2}{+}\cdots){+}(2{+}2^{n_1+1}{+}2^{n_1+n_2+1}{+}\cdots) = 2^2{+}2^{n_1}{+}2^{n_1+1}{+}2^{n_1+n_2}{+}{+}2^{n_1+n_2+1}{+}\cdots$$

Dividing by 4, we get $1 + 2^{n_1-2} + 2^{n_1-1} + 2^{n_1+n_2-2} + \cdots$, which in chain representation is:

$$\sim\ (n_1-2)\ \sim\ 1\ \sim\ (n_2-1)\ \sim\ 1 \sim \cdots$$

as we sought to show

The fact that the first n_1 behaves differently from the rest of the n_i's can be interpreted this way: when reading a chain from left-to-right to determine the next values of the chain, we always start in an *active state*, or an *active propagation*. If we encounter an $n \geq 3$ at the beginning, then n becomes $n-2$ and splits adding a 1, and the state becomes *inactive*. When in an inactive state, all $n \geq 3$ shall diminish by 1 to become $(n-1)$ as well as split (i.e. $\sim\ (n-1) \sim\ 1)$[1]. In this way, $n \geq 3$ in a

[1] In lemma 9, we shall see that the one that split can consume when an active propagation is triggered, hence it is

chain can be thought of as "unstable" and always shrinking.

All the proceeding proofs shall follow a very similar pattern to the above, and so the details of each will now be left out (the reader is free to verify at least a few of the next lemmas, as the intuition on the computation of these chains by reading the chain from left-to-right will be taken for granted in all of sections after this one). Furthermore, notice that in each of these calculations where always calculated from left-to-right, that is, the calculations of the next component was done after determine the value and behaviour of the immediately previous component. We shall henceforth say that the process of computing a chain and processing it's value by computing the values from left-to-right as the *propagation*.

Let us now mix n's and 2's. An interesting pattern to notice is that with the positioning of 2's in a chain. If it is in behind a digit $n \geq 3$ (to the left), then the 2 "behaves" like a digit greater than 2:

Lemma 6: Inactive 2

Let t be the number represented by the chain $\sim\ n\ \sim\ 2$ for $n \geq 3$. Then:

$$F(\sim\ n\ \sim\ 2) = \sim\ (n-2)\ \sim\ 1\ \sim\ (2-1)\ \sim\ 1 = \sim\ (n-2)\ \sim\ 1\ \sim\ 1\ \sim\ 1$$

Proof **:**
This is a matter of computation

The idea behind this lemma is that if a 2 is "stuck", then the power's cannot group, and hence each 2 shall split. Notice that this lemma can be generalized for sequence of 2 so that:

$$F(\sim\ n\ \sim\ 2 \sim\ \cdots\ \sim\ 2) = \sim\ (n-2)\ \sim\ 1\ \sim \cdots \sim\ 1$$

given that the 2's are not at the beginning of the chain (in which case we apply lemma 3).

Next, let us mix 1's and 2's. This behaviour is in fact the most nuanced behaviour of any propagation. Let us point out the simplest case of there being a sequence of ones followed by a sequence of two's before getting into all the possible mixed cases:

Lemma 7: 2 In front of 1's

Let t be the number represented by the chain $\sim\ 1\ \sim\ \cdots \sim\ 1\ \sim\ \underbrace{2\ \sim \cdots \sim\ 2}_{k \text{ times}}$. Then:

$$F(\sim\ 1\ \sim\ \cdots \sim\ 1\ \sim\ 2\ \sim \cdots \sim\ 2) = \underbrace{\sim 1 \cdots \sim 1}_{\text{one less}}\ \sim\ \underbrace{(2k+2)}_{\text{new digit}}$$

The idea is that the 2 that follow a 1 get *consumed*, and each following 2 shall also be consumed until the end is hit, at which point the value of the 2's will be *transmitted* at the end. The fact that we get $2k+2$ instead of $2k$ comes from the fact that the original 1 from the beginning of the chain must also get it's value transmitted, and that any active propagation will end by adding an extra value of 1 at the end of the active propagation (as shall be shown in the next lemma).

For the rest of the possible cases of having components with values either 1 or 2, they are listed out bellow with a brief explanation of each. These explanations shall motivate the intuition that a chain can be computed by reading it left to right:

possible that the 1 splits and gets immediately consumed

1. Consider first $\sim 1 \sim 1 \sim 2 \sim 1 \sim 1$. Then the first 1 is consumed to activate the propagation, then the 2nd one stays neutral, the first 2 gets consumed, then since this leaves a gap the next 1 gets modified and becomes a 3 (this should be interpreted as its value is transmitted at the next 1). The next 1 is unchanged, and a new 2 is added at the end, finishing the "active propagation". Ultimately, the change after applying F looks like:

$$\sim 1 \sim 1 \sim 2 \sim 3 \sim 1 \sim 2 \Rightarrow \sim 1 \sim 3 \sim 1 \sim 2$$

2. What if there is sequence of 2's followed by two ones. Then they shall all get absorbed into the one:

$$\sim 1 \sim 1 \underbrace{\sim 2 \sim \cdots \sim 2}_{k \text{ times}} \sim (2k+1) \sim 1 \sim 2 \Rightarrow \sim 1 \sim (2k+1) \sim 1 \sim 2$$

3. What if there is only one 1 following the 2's. Then that single one shall absorb the value of the 2's and a new 2 shall still occur:

$$\sim 1 \sim 1 \underbrace{\sim 2 \sim \cdots \sim 2}_{k \text{ times}} \sim (2k+1) \sim 2 \Rightarrow \sim 1 \sim (2k+1) \sim 2$$

4. What if there is a single 1 in front. Then it is the catalyst for a propagation, or it starts the active propagation:

$$\sim 1 \underbrace{\sim 2 \sim \cdots \sim 2}_{k \text{ times}} \sim (2k+1) \sim 2 \Rightarrow \sim (2k+1) \sim 2$$

5. What if there is no 1 following the 2's. Then we get the following behavior

$$\sim 1 \underbrace{\sim 2 \sim \cdots \sim 2}_{k \text{ times}} \sim (2k+2) \Rightarrow \sim (2k+2)$$

   In particular, the extra value within the propagation is absorbed with the 2's. This is exactly the content of lemma 7.

6. Finally, the last case to explore is if a sequence of 2's starts the chain and is followed by a sequence of 1's. Since there is no leading 1, there is no term to absorb the values of these and so they disappear:

$$\underbrace{\sim 2 \sim \cdots \sim 2}_{k \text{ times}} \sim 1 \sim 1 \sim 2 \Rightarrow \sim 1 \sim 2$$

   This is simply lemma 1 and lemma 3 combined. With our new found knowledge, we can say this is the behaviour of 2's in the *pre-active state* which shall become *active* if it encounter's a 1, or by lemma 4 will become inactive if it encounters an $n \geq 3$.

This gives a good idea for what happens when working exclusively with 1's and 2's. For example:

$$F(\sim 2 \sim 2 \sim 1 \sim 2 \sim 1 \sim 2) = \sim 3 \sim 4$$

Breaking down this example:

1. The first two 2's disappear as it is a pre-active propagation

2. first 1 activates the propagation hence gets consumed
3. The first 2 get absorbed into the next 1
4. Finally the propagation finishes, and as it is the end of the propagation it must transmit all values it has accumulated, which in this case is 4 as explained in lemma 7.

Lemma 8: Exclusively 1s And 2s

Let n be a number whose chain representation exclusively has 1's and 2's. Then after applying F:

1. Any two at the beginning is disappears from the chain
2. Any sequence of twos between ones will have its values absorbed and transmitted to the next one
3. Any sequence of two at the end (including a sequence of length zero) will "dump" it's value as well as add 2 to the total value

Let us now consider a chain with arbitrary values in each component. As we saw in lemma 5, how they transform through F seems to depend on their position: the first n_1 in the chain representation of lemma 5 diminished by 2 while all the other ones diminished by 1. Furthermore, by contrast to lemma 8, we saw that a 2 acts very differently when it's proceeded or preceded by an $n \geq 3$; in fact when it is preceded by an $n \geq 3$ it acts like a number ≥ 3 in the sense that it's value diminishes by 1 and splits just like the other numbers in lemma 5. Hence, the behavior of the 2 changes. This behavior is given a name:

Definition 3: Active and Inactive Elements

Let C be some arbitrary chain. Then

1. If after applying F, a 2 is being removed from the chain (whether it's valued is transmitted or removed), this 2 is said to be an *active* 2. A digit 2 is said to be *pre-active* if it is at the beginning of the chain, and there is either nothing or another pre-active2 preceding it.
2. If the two is splitting into $-\ 1\ -\ 1$, then it said to be an inactive 2.

More generally, call all elements in a chain *inactive* if they are either an inactive 2 or ≥ 3, and call all elements in a chain *active* if it is either an active 2 or 1.

As a consequence of all the above lemma's, it is impossible that a 2 at the beginning of a chain be inactive; a 2 is only inactive it is preceded by another inactive 2 or by an element $n \geq 3$, and a 2 is active if it is at the beginning of a chain, preceded by an active 2, or a 1. This means that 1's are in some sense that "catalyst" to activate 2. These last few lemmas make this precise, and covers the case of mixing 1's and n's:

Lemma 9: Case 1: $\sim n \sim 1 \sim 1 \sim p$

Let t be a number represented by a chain $C \sim\ n\ \sim\ 1\ \sim\ 1\ \sim\ p\ \sim\ D$ for some chains C, D, inactive n, and $p \geq 3$. Then:

$$F(C\ \sim\ n\ \sim\ 1\ \sim\ 1\ \sim\ p\ \sim\ D) = F(C)\ \sim\ (n-1)\ \sim\ 2\ \sim\ 2\ \sim\ (p-2)\ \sim\ \cdots$$

where p and D do not affect the behavior earlier in the chain:

$$F(C\ \sim\ n\ \sim\ 1\ \sim\ 1) = F(C)\ \sim\ (n-1)\ \sim\ 2\ \sim\ 2$$

and if C is null:

$$F(\sim\ n\ \sim\ 1\ \sim\ 1) = \sim\ (n-2)\ \sim\ 2\ \sim\ 2$$

Proof **:**
this is a matter of computation

Lemma 10: Case 2: $\sim n \sim 1 \sim \cdots \sim 1 \sim p$

Let t be a number represented by a chain $C \sim\ n\ \sim\ 1\ \cdots\ \sim\ 1\ \sim\ p\ \sim\ D$ for some chains C, D, inactive n, and $p \geq 3$. Then:

$$F(C\ \sim\ n\ \sim\ 1\ \cdots\ \sim\ 1\ \sim\ p\ \sim\ D) = F(C) \sim (n-1)\ \sim\ 2\ \sim\ 1\ \sim\ \cdots\ \sim\ 1\ \sim\ 2\ \sim\ (p-2)\ \sim\ \cdots$$

where p and D does not affect the behavior earlier in the chain:

$$F(C\ \sim\ n\ \sim\ 1\ \cdots\ \sim\ 1) = F(C) \sim (n-1)\ \sim\ 2\ \sim\ 1\ \sim\ \cdots\ \sim\ 1\ \sim\ 2$$

and if C is null

$$F(\sim\ n\ \sim\ 1\ \cdots\ \sim\ 1) = (n-2)\ \sim\ 2\ \sim\ 1\ \sim\ \cdots\ \sim\ 1\ \sim\ 2$$

Proof **:**
this is a matter of computation

When proving the above lemma, notice that the n splits, and the added 1 gets consumed as part of initializing the active propagation. Furthermore, the 1 proceeding the n (to the right of it) also gets consumed. This is different from the case where a chain started with 1's, for in that case there is no extra 1 consumed from the split that gets transmitted to the next 1. This behavior is why when starting a propagation, it is in a *pre-active state* or *pre-active propagation*, and requires the consumption of a non-split one to become active. This is new behavior the reader should keep in mind while looking at the different subtleties that may occur.

Lemma 11: Case 3: $\sim n \sim 1 \sim p$

Let t be a number represented by a chain $C \sim\ n\ \sim\ 1\ \sim\ p\ \sim\ D$ for some chains C, D, inactive n, and $p \geq 3$. Then:

$$F(C\ \sim\ n\ \sim\ 1\ \sim\ p\ \sim\ D) = F(C)\ \sim\ (n-1)\ \sim\ 3\ \sim\ (p-2)\ \sim\ \cdots$$

where p and D does not affect the behavior earlier in the chain:

$$F(C\ \sim\ n\ \sim\ 1) = F(C)\ \sim\ (n-1)\ \sim\ 3$$

and if C is null

$$F(\sim\ n\ \sim\ 1) = \sim\ (n-2)\ \sim\ 3$$

***Proof* :**
this is a matter of computation

Lemma 12: Case 4: $\sim 1 \sim n$

Let t be a number represented by a chain $\sim\ 1 \sim\ n\ \sim\ p\ \sim\ D$ for some chain D, $n \geq 3$, and inactive p Then:

$$F(\ \sim\ 1\ \sim\ n\ \sim\ p\ \sim\ D) =\ 2\ \sim\ (n-2)\ \sim\ (p-1)\ \sim\ \cdots$$

where p and D does not affect the behavior earlier in the chain:

$$F(\ \sim\ 1\ \sim\ n) =\ 2\ \sim\ (n-2)$$

***Proof* :**
this is a matter of computation

Lemma 13: Case 5: $\sim 1 \sim \cdots \sim 1 \sim m$

Let t be a number represented by a chain $\sim\ 1 \sim\ \cdots\ \sim\ n\ \sim\ p\ \sim\ D$ for some chain D, $n \geq 3$ and inactive p. Then:

$$F(\underbrace{\sim\ 1\ \cdots\ \sim\ 1}_{k \text{ times}}\ \sim\ n\ \sim\ p\ \sim\ D) = \underbrace{\sim\ 1\ \cdots\ \sim\ 1}_{k-1 \text{ times}}\ \sim\ 2\ \sim\ (n-2)\ \sim\ (p-1)\ \sim\ \cdots$$

where p and D does not affect the behavior earlier in the chain:

$$F(\underbrace{\sim\ 1\ \cdots\ \sim\ 1}_{k \text{ times}}\ \sim\ n) = \underbrace{\sim\ 1\ \cdots\ \sim\ 1}_{k-1 \text{ times}}\ \sim\ 2\ \sim\ (n-2)$$

***Proof* :**
this is a matter of computation

The 1's consistently diminish the value of an $n \geq 3$ by 2 in the same way an element n at the beginning of a chain gets it's value diminished by 2, and furthermore they are the key to activating 2's. With these lemmas, we shall make precise the terminology alluded to so far:

Definition 4: [Collatz Chain] Propagation: Pre-active, Active, and Inactive

Let C be a chain without leading 2's (due to lemma 3). Then the process to determine the value of $F(C)$ by reading the chain left-to-right to determine the next values is called reading the *propagation* of a chain. Each propagation starts at a *pre-active state* before the first number is considered. Then

- If the chain starts with an $n \geq 3$, then the propagation becomes an *inactive propagation*, or goes into an *inactive state*.
- If the chain starts with a 1, then it becomes an *active-propagation*, or goes into an *active state* and the 1 is said to be *consumed*.

Furthermore:

1. When a chain is in an activate state, 2's are activated.
2. An active-propagation *terminates* when it reaches an $n \geq 3$ or the *end* of the chain, at which the propagation becomes an inactive propagation and any 2's thereon are inactive. If a propagation terminates by reaching an $n \geq 3$, that n shall diminish its value by 2. All terminating propagation's shall end by adding a new elements which shall transmit all the accumulated value (if any) as well as add 2 to the total.
3. A propagation my reoccur if *catalyzed* by a 1 (consuming it in the process), at which point the propagation becomes an active propagation and any 2's thereon are active.
4. A propagation catalyzed after an inactive element shall transfer the initial value of the 1 to the next available 1. If no next available 1 exists then the value is added to the end of the active propagation.

Of all the possibilities we have covered, there is one combinations of 1's, 2's, and n's that we have yet to consider, namely when $\sim\ n\ \sim\ 1\ \sim\ 2\ \sim\ \cdots\ \sim\ 2\ \sim\ 1$. Covering this shall give the final possible arithmetical operation:

Lemma 14: Augmented 2

Let t be a number represented by a chain $C \sim n \sim 1 \sim 2 \sim \cdots \sim 2 \sim 1 \sim p$ where C is some chain, n is inactive and $p \geq 3$. Then:

$$F(C \sim n \sim 1 \underbrace{\sim 2 \sim \cdots \sim 2}_{k \text{ times}} \sim 1 \sim p) = F(C) \sim (n-1) \sim (2k+2) \sim 2 \sim (p-2)$$

where the C can as before be null and p has no effect on the earlier behavior. With color, this transformation can be represented as:

$$\sim (n - (1 \text{ or } 2)) \underbrace{\sim 1}_{\text{catalyst}} \underbrace{\sim 2 \sim \cdots \sim 2}_{\text{removed}} \underbrace{\sim (2k+2)}_{\text{absorbed 2's +1 from catalyst}} \underbrace{\sim 2}_{\text{propagation terminated}}$$

If the chain was of the form $n \sim 1 \sim 2 \sim \cdots \sim 2$ (or ending with a $p \geq 3$), then the propagation will look like:

$$\sim (n - (1 \text{ or } 2)) \underbrace{\sim 1}_{\text{catalyst}} \underbrace{\sim 2 \sim \cdots \sim 2}_{\text{removed}} \underbrace{\sim (2k+3)}_{\text{absorbed 2's, +1 from catalyst + 2 termination}}$$

This leads us to the final sub-state a propagation can be in: *augmented*.

Definition 5: [Collatz Chain] Propagation: Augmented

An *augmented active propagation* or an *augmented state* is a state triggered when an active propagation starts from an inactive element (an inactive 2 or $n \geq 3$). This state terminated either when:

1. The next 1 is encountered, in which case the 1 taken from the split n gets transmitted to it. It is possible that there is a sequence of 2's that preceded the one whose values also get transmitted
2. The next $n \geq 3$ is encountered. In which case, the augmented and active state both end (as shown in lemma 14).

Overall, a propagation can either be active or inactive, and an active propagation can further be augmented or not depending on its initialization. The only exception to these states is the pre-active state that is the state before the propagation starts and the propagation remains in this state for each 2 at the beginning of the chain.

From all of these lemmas, we can also conclude the following about the appearing of new digits and delegation of old digits:

Theorem 1: New and disappeared Digits

let $C \in \mathcal{C}$ be a chain. Then

1. The last digit of $F(C)$ is *always* a new digit.
2. More generally, given a digit $t_i = n$ of C where $n \geq 3$, then the digit before it will be a new digit.
3. New digits only appear before $t_i = n \geq 3$ and the end of the chain
4. Digits disappear if and only if:
   (a) They are a 1 in-front of the chain catalyzing the first active propagation
   (b) They are a 1 in-front of an n catalyzing an active propagation
   (c) They are a sequences of 2's in front of the chain
   (d) There is a sequence of 2's when the chain is in a state of active propagation

Proof **:**
By doing the manual computations in each of the above lemmas (1, 4, 5, 6, 7, 8, 9, 10, 12, 13), each new digit comes when we add the copy of the number at the end of a active propagation as shown, and similarly a digit i

It shall be important to keep track of new digits and disappeared digits when we calculate probabilities in the section 1.3.2. There shall usually be digits that remains. They will either keep their same value or change in value. For section 1.3.1, we will need to keep track of this information, and hence we box a definition for this:

Definition 6: Actions

Let $C_0 \in \mathcal{C}$ be a chain. Then at each digit $C_{0,i}$ of C_0, the *action* is the specific behaviour that will follow at the next step of the propagation. There are 5 types of actions that fall in 3 categories:

1. *Transformative actions*: actions that keep the same digit but may modify it in some way:
   (a) α_b^a-action: applies to an $n \geq 3$, it diminishing by 1 or 2
   (b) β_1^1-action: a 1 remaining unchanged
   (c) γ_b^a-action: a 1 increasing in value by a collapse transmitting its value
2. *creation action*: creates a new digit
   (a) δ_1-action: pairs with an α which does not trigger a new active propagation
   (b) $\delta_{2k+\ell}$-action: comes from the end of a collapse
3. *deletion action*: either permanently removes a digit or removes the digit so that its value is going to be transmitted to another digit:
   (a) ϵ^{φ}-action: only applicable to the 2' at the head of a chain
   (b) ϵ^a-action: removes 2's or 1's when applicable.

1.2 Quantifying chains

There are two natural functional that we can define on a chain, namely if $\mathcal{C}$ is the collection of all chains representing odd numbers, then:

$$\text{sum functional: } \sum : \mathcal{C} \to \mathbb{N},\ \sum(\sim n_1 \ \sim\ \cdots\ \sim\ n_k) = \sum_i n_i$$

$$\text{length functional: } m : \mathcal{C} \to \mathbb{N},\ m(\sim n_1 \ \sim\ \cdots\ \sim\ n_k) = k$$

The length functional is rather complicated when iterating F, and we shall return to it in section 1.3.2 as it has an interesting expected value. On the other hand, the first functional is remarkably simple when iterating F:

Theorem 2: Summing Properties

Let (C_i) be a Collatz sequence. Then:

1. If the chain starts with a 1 and ends with an active state, $\sum(C_{i+1}) = \sum(C_i) + 1$
2. If the chain starts with a 1 and ends with an inactive r, $\sum(C_{i+1}) = \sum(C_i)$
3. If the chain starts with an n and ends with an active state, $\sum(C_{i+1}) = \sum(C_i)$
4. If the chain starts with an $n \geq 3$ and ends with an inactive r, $\sum(C_{i+1}) = \sum(C_i) - 1$
5. If the chain starts with n number of 2's, then $\sum(C_{i+1}) = \sum(C_i) - 2n \pm [1/0]$ where the additional values added or subtracted (or no value added) depends on the 4 above conditions.

***Proof* :**
The 5th case is a consequence of lemma 3 and cases (1)-(4), hence we focus on those.

Let us first decompose the chain into blocks, each block starting with either an $n \geq 3$, an inactive element, or if neither one nor the other are available than the first element of the chain. A block starts at this element, and continues onto the next inactive element or $n \geq 3$. Examples of blocks in a chain are:

- The chain $\sim\ 1 \sim\ 1 \sim\ 1$ is composed of a single block
- The chain $\sim\ 1 \sim\ 2 \sim\ 1$ is composed of a single block
- The chain $\sim\ n \sim\ 1 \sim\ 2 \sim\ 1$ is composed of a single block
- The chain $\sim\ 1\ \sim\ n \sim\ 1 \sim\ 2 \sim\ 1$ is composed of two blocks, $[\sim\ 1]$ and $[\sim\ n \sim\ 1 \sim\ 2 \sim\ 1]$
- The chain $\sim\ 2\ \sim\ 2 \sim\ 1\ \sim\ n \sim\ 1 \sim\ 2 \sim\ 1\ \sim\ n\ \sim\ r$ is composed of 4 blocks, $[\sim\ 2\ \sim\ 2\ \sim\ 1]$, $[\sim\ n \sim\ 1 \sim\ 2 \sim\ 1]$, $[\sim\ n]$, and $[\sim\ r]$ where $r \geq 2$ is inactive.

Starting off with the simplest case of a chain of 1, by lemma 1 we get that

$$\sum(C_{i+1}) = \sum(C_i) + 1$$

If the chain started with an $n \geq 3$, then we see that after applying F we get that n becomes $n-2$, the $\sim\ 1$ that would be generated by n is absorbed into the propagation, while an extra value of 1 is added at the end of the propagation, balancing out to:

$$\sum(C_{i+1}) = \sum(C_i)$$

Similarly, if C_i was a chain of 1's that ended with an $r \geq 3$, then the extra value introduced by the propagation would be countered by the hit on n since $\sim\ n$ would become $\sim\ (n-2)\ \sim\ 1$ and hence:

$$\sum(C_{i+1}) = \sum(C_i)$$

Finally, if the starts and with an n with 1's in the middle, it never had the opportunity to add the extra 1 from the initial propagation. After applying F, The first block would have the same sum, while the last block would diminish in value, giving:

$$\sum(C_{i+1}) = \sum(C_i) - 1$$

Now, if we introduced active 2's in the chain, their values will be moved around in the chain, and hence not affecting the overall sum. Elements $n \geq 3$ that are not hit and are not contributing their 1 to initialize a propagation have their sum unchanged between a propagation, for example:

$$\sim\ n\ \sim \qquad \Rightarrow \qquad \sim\ (n-1)\ \sim\ 1\ \sim$$

If the one is contributed to the chain propagation, but the element is not hit, then we include the entire propagation in the block, and its total sum goes up by 1, however the next block will diminish its sum by 1 For example:

$$\sim\ 1 \sim\ 1 \sim\ 4 \sim\ 1 \sim\ 1 \sim\ 4 \sim\ 4 \sim\ 1 \sim\ 1$$
$$\Rightarrow$$
$$\sim\ 2\ \sim\ 2\ \sim\ 2\ \sim\ 2\ \sim\ 2\ \sim\ 2\ \sim\ 1\ \sim\ 3\ \sim\ 2\ \sim\ 2$$

Hence, every propagation adds a value, but if it hits another block that block diminishes in value, and if a block has no propagation and is not hit by a propagation its value is unchanged.

Finally, we may determine the effects of all the propagations simply by looking at the first and last element of the chain. If the first element is an $n \geq 3$, then there is no initial propagation, only a partial propagation, starting off the sum of C_{i+1} with a -1. Then every propagation would add a value of 1, but as long as it's followed up by another block the value will be countered. Hence, if the chain ends with an inactive element,

$$\sum(C_{i+1}) = \sum(C_i) - 1$$

If the chain ends with a 1 or active 2, then the value added by the last propagation cancels out the minus 1 value from the beginning, hence:

$$\sum(C_{i+1}) = \sum(C_i)$$

If the chain starts with a 1, then we get the same process except there is an additional value at to contend with, giving us the other two possibilities, completing the proof.

We box the notion of a blocks for future reference:

Definition 7: Block

Let C be a chain:

$$\sim\ t_1\ \sim\ t_2\ \sim\ t_3\ \sim\ \cdots$$

Then:

1. The *head block* is the digits starting from t_1 and ending at the first $t_i \geq 3$ for $i \geq 1$ (it is possible that the block contains only t_1).
2. An *active block* starts with (and includes) a $t_i \geq 3$ that is proceeded by a $t_{i+1} = 1$, and ends at the next $t_j \geq 3$ (and does *not* include t_j).
3. An *inactive block* is a single $t_i \geq 3$ that is not proceeded by a $t_{i+1} = 1$
4. the *tail block* is the last block of a chain.

The collective term for these 4 definitions is a *block*.

Thus, there is a surprising amount of rigidity over the size of a chain, it may only grow by at most 1, and it shrinks by 1 unless there is a sequence of two in-front of the chain that drastically diminish the size of a chain. As theorem 2 tells us, this size depends on the first and last digit and their states. These digits dynamically change as F iterates. If there were some regularity to these digits behaviour under F, we may be able to draw some interesting conclusion. Thus, we turn to probability where we shall see that these digits do behave in a mostly stable manner when considering the right probability distribution. The stability of the probability we shall introduce will also make working with the length functional $m(-)$ tractable.

1.3 Distribution of Head and Middle of Chain

Given an arbitrary chain $C \in \mathcal{C}$ before applying F, the digits within a chain are distributed rather nicely, that is given an odd $t \in \mathbb{N}_{>0}$ represented as a chain $\sim\ t_1\ \sim\ t_2\ \sim\ \cdots \sim\ t_k$:

$$P(t_i = 1) = 0.5 \qquad P(t_i = 2) = 0.25 \qquad P(t_i \geq 3) = 0.25$$

This shall be proven shortly. The fact shouldn't be a surprise as chain representation comes from counting in binary. What is more surprising is that this distribution remains the same for *most* digits while applying F. The goal of this section is to show just that. The digits for which this does not apply to is the *tail* of the chain, more particularity, it in general doesn't apply for about the last 5-10 digits of an arbitrary chain iterating through F, and these digits shall require special consideration.

Let us first establish a fact about the length of a chain. Given some $r \in \mathbb{N}$ consider the subset $\mathcal{C}_0^{\leq r} \subseteq \mathcal{C}$ of all chains given by odd digits less than r, we may ask what is the chance to have a chain of length k. This turns out to be normally distributed, as can be shown with a counting argument. For example, here is the distribution of the length of chains of $\mathcal{C}_0^{\leq (2^{21}-1)}$ (note by how chains were defined, even numbers do *not* have a chain representation, thus the aforementioned set has $\approx 1'000'000$ chains, not $\approx 2'000'000$):

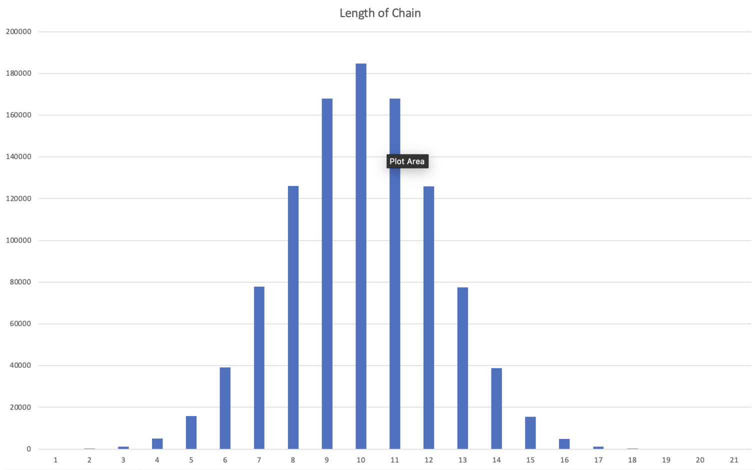

Figure 1.1: Length Distribution

It is a counting and convergence argument to see that as $r \to \infty$, this distribution remains normal, and hence when grabbing a random chain, we may always assume we are taking the chain from a set

$$\mathcal{C}_N = \{C \in \mathcal{C} \mid m(C) \leq N\}$$

for N large enough so that the probability that we have a number at the position we want (ex. the kth position) is ∼100%. In particular, as $N \to \infty$, we may assume that the probability of each digit of the chain is essentially independent of length consideration (i.e. pick N large enough so that $P(m(C) \leq k) \approx 0$ where $m(C)$ is the length of $C \in C_N$). Another way of looking at it is

$$\mathcal{C}^N = \{C \in \mathcal{C} \mid m(C) \geq N\}$$

excludes the (infinite) subset of $\mathcal{C}$ containing chains of length $> N$.

With the length of chain taken care of, let us calculate the probabilities of having either a 1, 2, or $n \geq 3$ at each position of a chain. An arbitrary chain can be represented as (for large enough N):

$$\sim \begin{bmatrix} 1 \\ 2 \\ n \end{bmatrix} \sim \begin{bmatrix} 1 \\ 2 \\ n \end{bmatrix} \sim \cdots$$

Lemma 15: Digit Distribution

Let $C \in \mathcal{C}_N$ be an arbitrary non-trivial chain of length $\leq N$ for some arbitrary large N. Consider the functionals $\pi_i(C) = n_i$ which returns the ith digit of the chain. Then as $N \to \infty$:

$$P(\pi_i(C) = 1) = 50\% \qquad P(\pi_i(C) = 2) = 25\% \qquad P(\pi_i(C) \geq 3) = 25\%$$

Proof **:**

Let C be an arbitrary chain representing the number t. Recall that a chain represents a number of the form:

$$1 + 2^{n_1}(1 + 2^{n_2}(1 + \cdots))$$

Then the probability question becomes: after subtracting 1, what is the probability of being able to divide n_1 times for some $n_1 \in \mathbb{N}$? This is in fact a uniform distribution argument; given any 2^k, half of the numbers are divisible by 2, a quarter divisible by 4, an eighth by 8, and so forth. Hence, there is a 50% chance $n_1 = 1$, a 25% chance that $n_1 = 2$, a 12.5% chance that $n_1 = 3$, and so froth. Summing up all the possibilities for $n_1 \geq 3$ gives us 25% chance. Hence:

$$P(\pi_1(C) = 1) = 50\% \qquad P(\pi_1(C) = 2) = 25\% \qquad P(\pi_1(C) \geq 3) = 25\%$$

Next, for π_2, we would be concerned whether there is a second digit, however as we have N large enough to not need to worry about the length, we may without consequence assume the required digits needed. Then the argument repeats itself verbatim and can be continued inductively, giving us:

$$P(\pi_i(C) = 1) = 50\% \qquad P(\pi_i(C) = 2) = 25\% \qquad P(\pi_i(C) \geq 3) = 25\%$$

as we sought to show.

Corollary 1: First Digit $n \geq 3$ after Collatz Function

Let $C \in \mathcal{C}_N$ be an arbitrary non-trivial chain of length $\leq N$ for some arbitrary large N. Assume $\pi_1(C) \geq 3$ Then as $N \to \infty$:

$$P(\pi_1(F(C)) = 1) = 50\% \qquad P(\pi_1(F(C)) = 2) = 25\% \qquad P(\pi_1(F(C)) \geq 3) = 25\%$$

Proof **:**

The first $\sim n \sim$ will become $\sim (n-2) \sim$, and may be 1 2, or ≥ 3. We need to calculate: given an $n \geq 3$, what is the probability of landing on a 1, 2, or ≥ 3. This can be figured out by finding the probability of n having value 3, 4, or $n \geq 5$, conditional on $n \geq 3$. By lemma 15, we know the distribution of all possible digits, namely every time we increase a number by 1, we half its possibility. Given that $n \geq 3$, we may sum up all the possibilities to get:

$$3 : 50\%, \qquad 4 : 25\%, \qquad \geq 5 : 25\%$$

as we sought to show.

Thus, there is a reasonable notion of a *normal chain*:

Definition 8: Normal Chain, Uniformity, and Variance

Let $\mathcal{C}$ be the collection of chains. Then a *normal chain* of length N is a chain that has a nonzero chance of being picked by choosing either $1, 2, 3, 4, ..., n, ...$ at each position with probability $50\%, 25\%, 12.5\%, ... \frac{1}{2^n}\%, ...$ respectively.

For sufficiently large N, a normal chain has digits $1, 2, n \geq 3$ distributed at a ratio $\approx$ $50\%, 25\%, 25\%$; such a normal chain is *uniform*.

If $C \in \mathcal{C}$ is a chain, and $\mathcal{D}(C) = \{1 : P_1, 2 : P_2 : n \geq 3 : P_3\}$ is a dictionary containing the percentage of digits $1, 2$ and ≥ 3 appearing in C, and $\mathcal{D}_E = \{1 : 0.5, 2 : 0.25,\ n \geq 3 : 0.25\}$, then the *variance* of C is:

$$\sigma(C) = \sum_{i=0}^{2} (\mathcal{D}(C)[i] - \mathcal{D}_E)$$

The notion of a normal chain will be important when considering the behaviour of the digits of a chain as it passes through the Collatz function F (the iterated chain shall no longer be a normal chain), and for infinitely long chains (the iterated chain shall be proven to remain normal)

1.3.1 Probability Invariance under F for Infinitely Long Chains

Let us now consider the harder case of looking at the behavior of the probability of each digit throughout the *entire* chain after passing through the Collatz function F. As it turns out, the behaviour at the head, middle, and tail of the chain shall be different. The head and the middle are almost identical (with the ~ 2 at the head needing to be considered) and are both very regular, while the tail is "chaotic". Due to this, we shall first prove the probabilistic distribution under the iteration of F at the beginning and middle of the chain. To do this, we shall start with the special case of *infinitely long chains*.

Definition 9: Infinite Chains

Let $\mathcal{C}_\infty$ denote the collection of infinitely long *normal chain*, namely all chains $\mathcal{C}_\infty$ can be chosen by choosing each digit $n_k \in \mathbb{N}$ with probability $1/2^m$ that the digit $n_k = m$ is chosen.

It is an exercise in local field theory that chains in C_∞ when converted into 2-adic representation and then into real numbers represent negative numbers (though not necessarily negative integers, or even rationals). Notice that Collatz Arithmetic is well-defined on infinitely long chain's as we may simply propagate indefinitely (in a similar way that arithmetic's is well-defined on the 2-adics). Note that there are chains that have a 0% probability of being picked, for example $\sim m \sim m \sim m \sim \cdots$.

It is also easy to see that Collatz conjecture is false for all 2-adic integers, as there are non-trivial cycles:

Example 1.1: Collatz False in 2-adics

Let

$$C = \sim 1 \sim\ 2 \sim\ 1 \sim\ 2 \sim\ 1 \sim\ 2 \cdots$$

This chain is periodic:

1. $C = \sim 1 \sim 2 \sim 1 \sim 2 \sim 1 \sim 2 \sim 1 \sim 2 \sim \cdots$

2. $F(C) = \sim 3 \sim 3 \sim 3 \sim 3 \sim 3 \sim 3 \sim 3 \sim 3 \sim \cdots$
3. $F^2(C) = \sim 1 \sim 1 \sim 2 \sim 1 \sim 2 \sim 1 \sim 2 \sim 1 \sim 2 \sim 1 \sim 2 \sim \cdots$
4. $F^3(C) = \sim 1 \sim 3 \sim 3 \sim 3 \sim 3 \sim 3 \sim 3 \sim 3 \sim \cdots$
5. $F^4(C) = \sim 2 \sim 1 \sim 1 \sim 2 \sim 1 \sim 2 \sim 1 \sim 2 \sim 1 \sim 2 \sim 1 \sim \cdots$
6. $F^5(C) = \sim 1 \sim 3 \sim 3 \sim 3 \sim 3 \sim 3 \sim 3 \sim 3 \sim \cdots$
7. $F^6(C) = \sim 2 \sim 1 \sim 1 \sim 2 \sim 1 \sim 2 \sim 1 \sim 2 \sim 1 \sim 2 \sim 1 \sim \cdots$

which repeats forever. Many more like these can be found, and there is certainly a paper that already exists about this as this is just examining the behaviour of the Collatz function on the 2-adics. [a]

There are lot of interesting questions here about whether most irrational number are normal chains, whether there are normal chains that have periods, but since we are interested in the case of finite chains we shall leave these questions open.

[a] there is: [Roz19]

Coming back to lemma 15 which proved the first digit probability distribution is invariant under F, generalizing to every position (over at least infinitely long chains) shall give us enough information understand how the probabilities (locally and globally) of the chain varies as we iterate F:

Theorem 3: Distribution – Infinitely Long

Let $C_0 \in C_\infty$ be an arbitrary infinitely long chain, and take the Collatz sequence $(C_0, C_1, ...)$. Consider the functionals $\pi_j(C_i) = n_{i,j}$ which returns the jth component of the chain C_i. Then

$$P(\pi_j(C_i) = 1) = 50\% \qquad P(\pi_j(C_i) = 2) = 25\% \qquad P(\pi_j(C_i) \geq 3) = 25\%$$

Proof **:**
Let $C_0 \in \mathcal{C}_\infty$, and take the Collatz sequence $(C_0, C_1, C_2, ...)$. Since C_0 has yet to have F applied to it, then by lemma 15 for all j

$$P(\pi_j(C_0) = 1) = 50\% \qquad P(\pi_j(C_0) = 2) = 25\% \qquad P(\pi_j(C_0) \geq 3) = 25\%$$

Let us now do induction on i and say that for all j

$$P(\pi_j(C_i) = 1) = 50\% \qquad P(\pi_j(C_i) = 2) = 25\% \qquad P(\pi_j(C_i) \geq 3) = 25\%$$

Let us now calculate the probabilities for the positions of C_{i+1}. What we shall do is manually calculate the first few positions of a general chain (i.e. the "head" of a general chain), and then show that any arbitrary element in the middle of the chain will have probability 0.5/0.25/0.25.

Starting with the case $\pi_1(C_i) = 1$, we know that by the rules of a propagation that the first one is "consumed", ostensibly it disappears. To calculate the probabilities of $\pi_1(C_{i+1})$, we need at least 3 digits for precise results:

%	25	12.5	12.5	12.5	6.25	6.25	12.5	6.25	6.25
C_i	$\sim 1 \sim 1 \sim 1 \sim$	$\sim 1 \sim 2 \sim 1 \sim$	$\sim 1 \sim n \sim 1 \sim$	$\sim 1 \sim 1 \sim 2 \sim$	$\sim 1 \sim 2 \sim 2 \sim$	$\sim 1 \sim n \sim n \sim$	$\sim 1 \sim 1 \sim n \sim$	$\sim 1 \sim 2 \sim n \sim$	$\sim 1 \sim n \sim n' \sim$
C_{i+1}	$\sim 1 \sim 1 \sim$	$\sim 3 \sim$	$\sim 2 \sim m \sim$	$\sim 1 \sim n \sim$	$\sim n \sim$	$\sim 2 \sim n \sim 1 \sim$	$\sim 1 \sim 2 \sim m \sim$	$\sim 4 \sim m \sim$	$\sim 2 \sim m \sim 1 \sim m' \sim$

Summing up, we get:

$$\begin{aligned} P(\pi_1(C_{i+1}) = 1 \,|\, \pi_1(C_i) = 1) &= 50\% \\ P(\pi_1(C_{i+1}) = 2 \,|\, \pi_1(C_i) = 1) &= 25\% \\ P(\pi_1(C_{i+1}) \geq 3 \,|\, \pi_1(C_i) = 1) &= 25\% \end{aligned}$$

Let's next consider the probabilities when $\pi_1(C_i) \geq 3$. The first $\sim n \sim$ will become $\sim (n-2) \sim$, and may be 1 2, or ≥ 3. We need to calculate: given an $n \geq 3$, what is the probability of landing on a 1, 2, or ≥ 3. This can be figured out by finding the probability of n having value 3, 4, or $n \geq 5$, conditional on $n \geq 3$, which by corollary 1:

$$3 : 50\%, \qquad 4 : 25\%, \qquad \geq 5 : 25\%$$

Thus, we get:

$$\begin{aligned} P(\pi_1(C_1) = 1 \,|\, \pi_1(C_0) \geq 3) &= 50\% \\ P(\pi_1(C_1) = 2 \,|\, \pi_1(C_0) \geq 3) &= 25\% \\ P(\pi_1(C_1) \geq 3 \,|\, \pi_1(C_0) \geq 3) &= 25\% \end{aligned}$$

For the probabilities if $\pi_1(C_i) = 2$, we know that by the rules of a propagation that the 2 will disappear. If there is any other 2 to the right of it, then that 2 will disappear as well. Hence, the possible first values of C_i are:

$$\begin{aligned} &[\sim 2 \sim] \sim 1 \sim 1 \sim && 25\% \\ &[\sim 2 \sim] \sim n \sim 1 \sim && 12.5\% \\ &[\sim 2 \sim] \sim 1 \sim 2 \sim && 12.5\% \\ &[\sim 2 \sim] \sim n \sim 2 \sim && 6.25\% \\ &[\sim 2 \sim] \sim 1 \sim n \sim && 12.5\% \\ &[\sim 2 \sim] \sim n \sim n' \sim && 6.25\% \end{aligned}$$

then given $m = n - 2$, the resulting value after the propagation is:

$$\begin{aligned} &{\color{blue}\sim 1 \sim} \\ &\sim m \sim \\ &{\color{green}\sim m \sim} \\ &\sim m \sim \\ &{\color{red}\sim 2 \sim} \\ &\sim m \sim 1 \sim m' \sim \end{aligned}$$

The blue marks chains where the 2nd digit is 1, the red where the 2nd digit is 2, the green where the 2nd digit has a chance of being greater than or equal to 3, and black is one of these three. Visually, we may see these as:

$$\begin{aligned} 25\% &: \sim 1 \sim \\ 12.5\% &: \sim 2 \sim \\ 12.5\% &: \sim n \sim \\ 25\% &: \{1, 2, n\} \end{aligned}$$

To determine the last 25%, we use our earlier calculations to redistribute the probabilities. Hence, appropriately re-distributing the probabilities we get:

$$P(\pi_1(C_1) = 1 \,|\, \pi_1(C_0) = 2) = 50\%$$
$$P(\pi_1(C_1) = 2 \,|\, \pi_1(C_0) = 2) = 25\%$$
$$P(\pi_1(C_1) \geq 3 \,|\, \pi_1(C_0) = 2) = 25\%$$

Though we have done this calculation manually, we could have also combined the previous two cases to

Combining all these results, we get:

$$P(\pi_1(C_1) = 1) = 50\%$$
$$P(\pi_1(C_1) = 2) = 25\%$$
$$P(\pi_1(C_1) \geq 3) = 25\%$$

We shall now show that the probabilities $P(\pi_i(C_{i+1}) = 1)$, $P(\pi_i(C_{i+1}) = 2)$, and $P(\pi_i(C_{i+1}) \geq 3)$ is exactly the same as for C_i, that is we keep the same type of normalness throughout the propagation!

We shall split up the case-work depending on the value of $\pi_1(C_i)$. Let's consider $\pi_1(C_i) \geq 3$ as it is the simplest. Then as always, $\pi_2(C_{i+1})$ on the first 3 digits of C_i. Let's split these into 9 cases representing the possibilities of the beginning of C_i:

$\sim n \sim 1 \sim 1 \sim$	25%
$\sim n \sim 2 \sim 1 \sim$	12.5%
$\sim n \sim n' \sim 1 \sim$	12.5%
$\sim n \sim 1 \sim 2 \sim$	12.5%
$\sim n \sim 2 \sim 2 \sim$	6.125%
$\sim n \sim n' \sim 2 \sim$	6.125%
$\sim n \sim 1 \sim n' \sim$	12.5%
$\sim n \sim 2 \sim n' \sim$	6.125%
$\sim n \sim n' \sim n'' \sim$	6.125%

Then given $m = (n - 2)$ and m', m'' is the resulting value after the propagation:

$$\sim m \sim 2 \sim$$
$$\sim m \sim 1 \sim 1 \sim$$
$$\sim m \sim 1 \sim m' \sim$$
$$\sim m \sim m' \sim$$
$$\sim m \sim 1 \sim 1 \sim 1 \sim 1 \sim$$
$$\sim m \sim 1 \sim m' \sim 1 \sim 1 \sim$$
$$\sim m \sim 3 \sim m' \sim$$
$$\sim m \sim 1 \sim 1 \sim 1 \sim m' \sim$$
$$\sim m \sim 1 \sim m' \sim 1 \sim m'' \sim$$

The blue marks chains where the 2nd digit is 1, the red where the 2nd digit is 2, the green where the 2nd digit has a chance of being greater than or equal to 3. Summing all the probabilities, we get:

$$P(\pi_2(C_1) = 1 \,|\, \pi_1(C_0) \geq 3) = 50\%$$
$$P(\pi_2(C_1) = 2 \,|\, \pi_1(C_0) \geq 3) = 25\%$$
$$P(\pi_2(C_1) \geq 3 \,|\, \pi_1(C_0) \geq 3) = 25\%$$

Let us next do the case of $\pi_2(C_0) = 1$. This case requires a good amount more work, which can be summarized with these image:

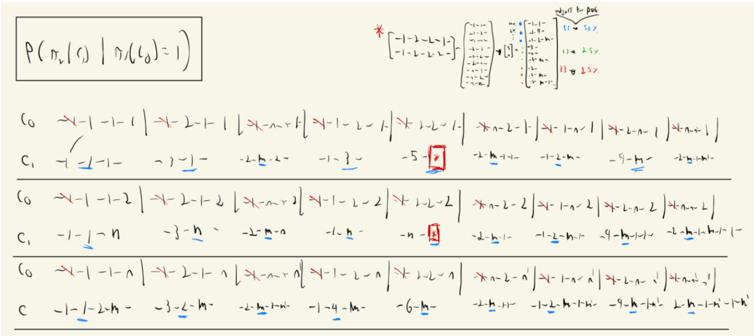

Figure 1.2: Calculations for possible 2nd digit given a length of 4

The probability calculations are given by this table:

probabilities counted up in binary						2ND DIGIT	1	2	n	m	X	
1	1	1	1	0.125	111	1	0.125					
1	0.5	1	1	0.0625	211	1	0.0625					
1	0.5	1	1	0.0625	n11	m				0.0625		
1	1	0.5	1	0.0625	121	3			0.0625			
1	0.5	0.5	1	0.03125	221	X					0.03125	
1	0.5	0.5	1	0.03125	n21	m				0.03125		
1	1	0.5	1	0.0625	1n1	2		0.0625				
1	0.5	0.5	1	0.03125	2n1	m				0.03125		
1	0.5	0.5	1	0.03125	nn1	m				0.03125		
1	1	1	0.5	0.0625	112	1	0.0625					
1	0.5	1	0.5	0.03125	212	n			0.03125			
1	0.5	1	0.5	0.03125	n12	m				0.03125		
1	1	0.5	0.5	0.03125	122	n			0.03125			
1	0.5	0.5	0.5	0.015625	222	X					0.015625	
1	0.5	0.5	0.5	0.015625	n22	m				0.015625		
1	1	0.5	0.5	0.03125	1n2	2		0.03125				
1	0.5	0.5	0.5	0.015625	2n2	m				0.015625		
1	0.5	0.5	0.5	0.015625	nn2	m				0.015625		
1	1	1	0.5	0.0625	11n	1	0.0625					
1	0.5	1	0.5	0.03125	21n	2		0.03125				
1	0.5	1	0.5	0.03125	n1n	m				0.03125		
1	1	0.5	0.5	0.03125	12n	4			0.03125			
1	0.5	0.5	0.5	0.015625	22n	m				0.015625		
1	0.5	0.5	0.5	0.015625	n2n	m				0.015625		
1	1	0.5	0.5	0.03125	1nn	2		0.03125				
1	0.5	0.5	0.5	0.015625	2nn	m				0.015625		
1	0.5	0.5	0.5	0.015625	nnn	m				0.015625		FINAL TALLY
				1			0.3125	0.15625	0.15625	0.328125	0.046875	1
						accounting for M	0.4765625	0.2382813	0.2382813			
						Accounting for X	0.5	0.25	0.25			

Figure 1.3: calculations for all probabilities

Which we see gives us:

$$P(\pi_2(C_{i+1}) = 1 \,|\, \pi_1(C_i) = 1) = 50\%$$
$$P(\pi_2(C_{i+1}) = 2 \,|\, \pi_1(C_i) = 1) = 25\%$$
$$P(\pi_2(C_{i+1}) \geq 3 \,|\, \pi_1(C_i) = 1) = 25\%$$

Finally, for $\pi_1(C_i) = 2$, this is a combination of the other two cases since by lemma 3, the 2's will disappear and reduces to the case of 1 or n that we have just covered, which gives us:

$$P(\pi_2(C_{i+1}) = 1 \,|\, \pi_1(C_i) = 2) = 50\%$$
$$P(\pi_2(C_{i+1}) = 2 \,|\, \pi_1(C_i) = 2) = 25\%$$
$$P(\pi_2(C_{i+1}) \geq 3 \,|\, \pi_1(C_i) = 2) = 25\%$$

Combining all the results gives:

$$P(\pi_2(C_{i+1}) = 1) = 50\%$$
$$P(\pi_2(C_{i+1}) = 2) = 25\%$$
$$P(\pi_2(C_{i+1}) \geq 3) = 25\%$$

We shall repeat this argument for the head digits via an inductive argument in the following proof (hence, we are also inductively assuming the distributions are accurate to the left of each digits). What is important is that the probabilities of the components at the head are the same for any state that we may be in.

Now, with probabilities of the digits of the head of the chain after applying F are calculated, let us now consider being in the middle of the chain. To do so, use the vocabulary introduced in definition 6 to keep track of the behaviour of digits. In particular, the digits in the chain may have multiple different behaviour depending on what digits that precede them. It is thus practical to have more granular control over which actions are happening on each digit:

1. $\alpha^n_{n-(1/2)}$: This represents the value of an inactive element with value n becoming $n-1$ or $n-2$ after applying F, for example given the chain $\sim\ n_a\ \sim\ n_b$ with $n_a, n_b \geq 3$, $F(\sim n_a \sim n_b)$ will create $(n_a - 2)$ and $(n_b - 1)$ which will have actions $\alpha^{n_a}_{n_a-2}$ and $\alpha^{n_b}_{n_b-1}$
2. β: this represents 1's which did not chain their values. In $F(\sim\ 1\ \sim\ 1)$, the 2nd one would have action β as it remains unchanged
3. $\gamma^{1^*}_2$: represents a 1 that changed values in the case of $\sim\ i\ \sim\ 1\ \sim\ 1$ where i represents an inactive element.
4. $\gamma^{2,k[^*]}_{2k+1/2}$: represents a 1 that changed values. For example in $F(\sim 5 \sim 1 \sim 2 \sim 1) =\sim 3 \sim 4 \sim 2$, the 3rd digit, 2nd two, went from a 1 to a 2 (do the proof of lemma 9 if more motivation is needed). The star, *, in the superscript represents a 1 that has had an augmented propagation effect its value.
5. δ_1: represents a new element that comes if the 1 created by an $n \geq 3$ is not consumed. For example, $F(\sim n_a \sim n_b)$ at least one 1 since n_a is followed by $n_b \geq 3$.
6. δ_{2k+2}: represents creating a new value at the end of a propagation that is not augment
7. δ^*_{2k+3} represents creating a new value at the end of a propagation that *is* augmented.
8. $\epsilon^{1,*}$: represents the one that is consumed to start a new propagation that is *not* the first digit of the chain. This explains why it is augmented.
9. $\epsilon^{2,k}$: represents consumed 2 that are not augmented.
10. $\epsilon^{2,k*}$: represents consumed 2 that *are* augmented.

11. ϵ^1: represents the first consumed 1
12. $\epsilon^{2,\varphi}$ Represents consumed two at the beginning of the chain. These are special as they completely disappear from the chain, and hence there is the φ symbol to represent this.

As we are working in the middle of the chain, it is not necessary to consider the last two possibilities as they only occur at the beginning. Using these, we may completely determine the value of a digit t_i in a chain given the digits and that precede it as well as which action is happening on those digits.

What we shall do now is calculate the following probability: given a digit t_i in a chain that is either 1, 2, or $n \geq 3$, what is the probability that the next digit after applying F will be 1, 2, n, or be part of a propagation (being part of a propagation means that the digit had an ϵ-action, meaning it was consumed). Each case shall be considered separately. This shall be a local analysis, in particular we are determining the probability of an individual digit given the behaviour of what is happening around the digit (in particular what's happening before the digit). This shall almost be enough to calculate the distribution of the digits, after having calculated the local behaviour we shall turn to a global perspective to finish off the calculations.

Starting with $t_i = 1$, we get the following table:

-1-	necessary circumstances	probabilities	result
alpha^n_{n-1/2}	(i2/n)-1	0.33333333	collapse
beta	(1/a2)-1-1-1	0.16666667	1
gamma^{1*}_2	-(n/i2)-1-1-1	0.08333333	1
gamma^{2,k*}_{2k+2}	-(n/i2)-1-2-[...]-2-1-1	0.02777778	1
gamma^{2,k}_{2k+1}	-1-1-2-[...]-2-1-1	0.04166667	1
^	-(a2)-1-2-[...]-2-1-1	0.01388889	1
epsilon^{1,*}	-n-1-1 and -n-2-[...]-2-1-1	0.16666667	2
epsilon^{2,k}	-1-1-2-[...]-2-1	0.08333333	n
^	(a2)-1-2-[...]-2-1	0.02777778	n
e^{2,k*}	-(n/i2)-1-2-[...]-2-1	0.05555556	n
	total	1	
P (digit's next value | digit is 1)			

Figure 1.4: P(digit's next value $\mid$ digit is 1)

Most of the calculations are a continuation of what was done before. The new circumstance to consider are the chains that have $2 \sim [...] \sim 2$. The [...] represents all possible chains composed of 2's (ex, $\sim$, $\sim 2 \sim$, $\sim 2 \sim 2 \sim$, and so forth). This is because we are considering *all* possible values before the chain (this is also why we had to show that the probability at the front of the chain is still invariant under F separately). If we had infinitely many digits that precede t_i (to the left of it), then the sum of all possible chains in [...] would be (1/3). In practice this sum is finite, and our induction hypothesis gives us the sufficient and necessary conditions to calculate these values. Note, that the longer the chain of twos, the less likely it is to appear, namely for each extra ~ 2 added the probability of the chain diminishes by 1/4. In fact, within "[...]", the chances of a chain of 2' being length ≥ 5 is $< 0.1\%$. Note that in the first row, the 1 has action ϵ^1 applied to it and hence is consumed. We shall see how to deal with consumed digits soon (in the figures, this is labeled as "collapse").

Summing the probabilities we get:

	prob's
1	0.33333333
2	0.16666667
n	0.16666667
collapse	0.33333333

Figure 1.5: Summing probabilities from figure 1.4

Let us next do the same thing but taking the case of $t_i = 2$. Following the same procedure where we look at the possible digits before t_i and consider what possible actions may occur, we get:

-2-	necessary circumstances	prob	result
alpha^n_{n-(1/2), 1 or nothing}	(inactive 2/n)-2	0.33333333	1
beta	-(active 2/-1)-1-1-2	0.16666667	collapse
gamma^{1,*}_2	-(inactive 2/n)-1-1-2	0.08333333	collapse
gamma^{2,k}_{2k+1}	-(active 2/-1)-1-2[...]-2-1-2	0.05555556	collapse
gamma^{2,k*}_{2k+2}	-(inactive 2/n)-1-2[...]-2-1-2	0.02777778	collapse
epsilon^{1,*}	-(inactive 2/n)-1-2	0.16666667	collapse
epsilon^{2,k}	-(active 2/1)-1-2[...]2-2	0.11111111	collapse
epsilon^{2,k^*}	-(inactive 2/n)-1-2[...]-2-2	0.05555556	collapse
	total:	1	
P (digit's next value | digit is 2)			

Figure 1.6: P(digit's next value $\mid$ digit is 2)

Summing the probabilities we get:

1	0.33333333
2	0
n	0
collapse	0.66666667

Figure 1.7: Summing probabilities from figure 1.6

The final case is where $t_i \geq 3$. In this case, we get:

-n-	concrete	prob	result
alpha^n_{n-(1/2), 1 or nothing}	(inactive 2/n)-n	0.33333333	(n-1)
delta_2	-(1/active 2)-1-n	0.33333333	(n-2)
delta_{2k+2}	(-1/active 2)-1-2[...]-2-n	0.11111111	(n-2)
delta^*_{2k+3}	-(n/inactive 2)-1[-2-...-2]-n	0.22222222	(n-2)
	total:	1	

Figure 1.8: P(digit's next value $\mid$ digit is ≥ 3)

Building off of the proof of corollary 1 it can be showed that:

$$P(n = 3|n \geq 3) = 0.5 \qquad P(n = 4|n \geq 3) = 0.25 \qquad P(n \geq 5|n \geq 3) = 0.25$$

From this, we may calculate the following:

(inactive 2/n)-n	0.33333333	(n-1)
2	0.16666667	
3	0.08333333	
n \ge 3	0.08333333	0.33333333
...		
-(1/active 2)-1-n	0.33333333	(n-2)
1	0.16666667	
2	0.08333333	
n \ge 3	0.08333333	0.33333333
...		
(-1/active 2)-1-2[...]-2-n	0.11111111	(n-2)
1	0.05555556	
2	0.02777778	
n \ge 3	0.02777778	0.11111111
...		
-(n/inactive 2)-1[-2-...-2]-n	0.22222222	(n-2)
1	0.11111111	
2	0.05555556	
n \ge 3	0.05555556	0.22222222

Figure 1.9: breaking down the probabilities in figure 1.8

Summing, this gives the result:

RESULTS	
1	0.33333333
2	0.33333333
n	0.33333333

Figure 1.10: Summing probabilities from figure 1.9

Putting the probabilities for the cases of $1, 2, n \geq 3$ all of together and correctly weighting the probabilities for each case, we get that the probability of the next digit is:

-1- -> -1-	0.3333333	0.1666667
-1- -> -2-	0.1666667	0.0833333
-1- -> -n-	0.1666667	0.0833333
-1- -> collapse	0.3333333	0.1666667
-2- -> -1-	0.3333333	0.0833333
-2- -> -2-	0	0
-2- -> -n-	0	0
-2- -> collapse	0.6666667	0.1666667
-n- -> -1-	0.3333333	0.0833333
-n- -> -2-	0.3333333	0.0833333
-n- -> -n-	0.3333333	0.0833333

Figure 1.11: adjusting probabilities of previous calculations

P(next = 1) =	0.3333333
P(next = 2) =	0.1666667
P(next \ge 3) =	0.1666667
P(next=col) =	0.3333333

Figure 1.12: Taking adjusted probabilities from figure 1.11 and summing

This gives us the local behaviour of the chain. Notice that the local behaviour does not take into account δ_1 (i.e. consumed digits), namely because this is an element that does not come from previous elements, and furthermore it does not allow us to "resolve" propagations. To solve these problems, we require taking a global perspective.

Let us first calculate the probability of δ_1. This is the probability that we have an inactive element followed by a 2 or $n \geq 3$, The probability of an inactive element is $1/3$ (the probability of $n \geq 3$, plus the probability of $\sim n \sim 2$, plus $\sim n \sim 2 \sim 2$, and so forth). The probability of an n or 2 is 0.5, and hence

$$P(\sim i \sim (n/2)) = 1/6$$

This means that if we were to pick an arbitrary element in an arbitrarily long chain, the chances that it is a 1 coming from a δ_1 is $1/6$. Next, let us calculate the probability of there being a propagation that results in a δ_k $(k \geq 2)$, since the results from a γ_b^a are already factored into the local probability analysis above (see the rows for $\sim\ 2[...]2\ \sim\ 1$ in figure 1.4), we only need to consider:

$$i \sim 1 \sim n,\ i \sim 1 \sim 2 \sim n,\ ... \qquad \text{or} \qquad a \sim 1 \sim n,\ a \sim 1 \sim 2 \sim 2 \sim n,\ ...$$

where i represents an inactive component, and a represents an active component. A quick calculations shows that the probability of i is $1/3$ and a is $2/3$[a]. Calculating we get that:

$$P(\text{consumed}) = 1/6$$

Finally, we may look at our above table to determine that the value after an active propagation terminates is:

$$P(\text{value} = 1|\text{propagation ends, new digit}) = 0$$
$$P(\text{value} = 2|\text{propagation ends, new digit}) = 0.5$$
$$P(\text{value} \geq 3|\text{propagation ends, new digit}) = 0.5$$

This leaves us with the chances that we have the same digit being 2/3 which are distributed 0.5/0.25/0.25. Putting together all the possibilities, we get:

same digit	collapse	delta_1	
0.6666667	0.16666667	0.1666667	THIS!
0.5	0	1	0.5
0.25	0.5	0	0.25
0.25	0.5	0	0.25

Figure 1.13: final results for local probability + global probability

Finally, to show that we get back the same probability distribution for each $n \geq 3$ (which is necessary for the computations of the next iteration), we require showing that:

$$P(n = 3|n \geq 3) = 0.5 \qquad P(n = 4|n \geq 3) = 0.25 \qquad P(n = 5|n \geq 3) = 0.125\cdots$$

Here are some of the calculations for these probabilities:

	beta	gamma*	delta*	gamma*	delta*	gamma*	delta*	gamma*	delta*	gamma*	delta*
(i2/n)=ん	-あ-1-1	-ん-1-1	-ん-1-n	-ん-1-2-1	-ん-1-2-n	-ん-1-2-2-1	-ん-1-2-2-n	-ん-1-2-2-2-1	-ん-1-2-2-2-n	-ん-1-2-2-2-2-n	...
	1	2	3	4	5	6	7	8	9	10	11
	0.333333333	0.166666667	0.083333333	0.041666667	0.020833333	0.010416667	0.005208333	0.002604167	0.001302083	0.000651042	0.000325521
		0.166666667	0.083333333	0.041666667	0.020833333	0.010416667	0.005208333	0.002604167	0.001302083	0.000651042	0.000325521
	1	2	3	4	5	6	7	8	9	10	11
(a2/1)=あ	-あ-1-1	-あ-1-n	-あ-1-2-1	-あ-1-2-n	-あ-1-2-2-1	-あ-1-2-2-n	-あ-1-2-2-2-1	-あ-1-2-2-2-n	-あ-1-2-2-2-2-1	-あ-1-2-2-2-2-n	...
		delta	gamma	delta	gamma	delta	gamma	delta	gamma	delta	gamma
sum is:	1	2	3	4	5	6	7	8	9	10	11
total:	0.333333333	0.333333333	0.166666667	0.083333333	0.041666667	0.020833333	0.010416667	0.005208333	0.002604167	0.001302083	0.000651042
gamma/delta (<-*)		0.166666667	0.083333333	0.041666667	0.020833333	0.010416667	0.005208333	0.002604167	0.001302083	0.000651042	0.000325521
adjusted for guaranteed num change:	0	0.5	0.25	0.125	0.0625	0.03125	0.015625	0.0078125	0.00390625	0.001953125	0.000976563
¬2 g/d	0	0.25	0.125	0.0625	0.03125	0.015625	0.0078125	0.00390625	0.001953125	0.000976563	0.000488281

Figure 1.14: Part of the calculations

Summing up, we get that for all j:

$$P(\pi_j(C_{i+1}) = k) = \frac{1}{2^k}\%$$

which completes the inductive argument, and hence the proof.

[a] For i, consider $P(n) + P(\sim n \sim 2) + P(n \sim 2 \sim 2)$... and for a consider $P(1) + P(\sim 1 \sim 2) + P(1 \sim 2 \sim 2) + ...$

From the proof, we see another interesting result. When computing the probability, we keep track of what happens *per digit*, and also what happens *per chain*. In particular, the transition actions give us per-digit probabilistic computation and had to include the probability of getting consumed, while the creation and delegation actions give us per-chain probabilistic computations. These result amalgamate to give the total probability at an arbitrary position, but it will be important to consider them separately when looking at finite chains:

Definition 10: Local and Global Probability

let C be a chain and C_i a digit of C. Then

1. the *local probability* at C_i is the probability that C_i becomes $1, 2, n \geq 3$, or consumed.
2. the *global probability* is the probability of given to collapsed values δ_k's and δ_1's.
3. the *total probability* is the sum of the local probability of a given digit and the global probability, given the total probability of the position (the i**th** digit)

We can immediately conclude the following about the local behaviour for finite chains:

Corollary 2: Probability of Same Digit is Invariant in Finite Chains

Let $C_0 \in \mathcal{C}$ be a [finite] normal chain, and let $C_{0,i}$ be a digit of C_0. Then if we are only keeping track of C_i during its life-span (i.e. until it get's consumed or removed after iteratively applying $F^{\circ k}$), Then its probability of the digit having the value $1, 2, n \geq 3$ is

$$50\% \qquad 25\% \qquad 25\%$$

Proof **:**
This is immediate from figure 1.13; it is the conditional probability under "same digit'.

Note how it is important to keep track of the digit to make this result work; the total probability at the end of a [finite] normal chain after 1 iteration is:

$$P(\pi_{-1}(F(C_0)|\pi_{-1}(C_0) = 1) = \frac{1}{3}$$
$$P(\pi_{-1}(F(C_0)\pi_{-1}(C_0) = 2) = \frac{1}{3}$$
$$P(\pi_{-1}(F(C_0)\pi_{-1}(C_0) \geq 3) = \frac{1}{3}$$

And this is simple to compute:

break-down	probability	value
P(-n|)	0.25	1
P(-i-2|)	0.0833333	1
P(-a-2|)	0.1666667	n
P(-i-1|)	0.1666667	3
P(-a-1|)	0.3333333	2
P(pi_{-1}=1)	0.3333333	
P(pi_{-1}=2)	0.3333333	
P(pi_{-1}>=1)	0.3333333	

Figure 1.15: last digit probability breakdown. The " | " represents end of chain

Notice that *all* of these digits are obtains from a δ, as already explained by theorem 1. These results also line up with figure 1.12 with the conditional probability that there is an n at the end of the partial chain, for the behaviour at the end of the sequence is the same as the active propagation hitting an n.

Thus, the end of the chain has its own probability calculations. This probability is *not* invariant under F, and so shall be explored further in section 1.4.

1.3.2 Consequences for Infinite Chains

As the probability of each digit is identical across a chain, we get the following global uniformization result:

Corollary 3: Uniformization of Infinite Chain

Let $C_0 \in \mathcal{C}_\infty$ be an infinite chain, and consider its Collatz sequence $(C_0, C_1, ..., C_n, ...)$. Take the sets $D_{i,j}$ which keep track of how many 1's, 2's and $n \geq 3$'s there are at position j of the chain at the ith position of the Collatz Sequence. For example, it is possible that $D_{2,3} = \{1 : 2,\ 2 : 1,\ n : 0\}$. which means that in 3rd position for C_0, C_1, C_2, there were two 1's, one 2, and no $n \geq 3$.

Then as $n \to \infty$, the ratio between these three quantities approach

$$0.5 \qquad 0.25 \qquad 0.25$$

This is called the "uniformization" as it says that given enough iterations, will have $1/2/n$ distributed uniformally

Proof **:**
This result is immediate from theorem 3.

We can in fact do one better: we can calculate the speed of at which these ratios converges to $0.5, 0.25, 0.25$, and the speed at which this happens is rather quick and dependent on the variance (though I did not give a formal proof of it, it should just be some known probability theory). We can also conclude that what moves a digits probability *away* from the variance is the pretense of a large digits; in fact for finite chains we can interpret the end of the chain as an arbitrarily large digit N (namely, the end of the chain acts identically to the case of there being an arbitrarily large digit N).

Due to theorem 3, we can make sense of the average *difference in size* and *difference in length* between two infinitely long chains even though the notion of size and length is not well-defined for an infinite chain:

Definition 11: Difference in Size, Difference in Length

let C be a (either finite or infinite, not necessarily normal) chain. Then:

1. If C is finite,

$$\Delta_m(C) = m(F(C)) - m(C) \qquad \text{and} \qquad \Delta_\Sigma(C) = \sum(F(C)) - \sum(C)$$

2. If C is infinite, then:

   (a) First create a list of actions on C, label it $A'(C)$.

   (b) Transform the list as follows:

      i. For $A_m(C)$: α becomes 0, β becomes 0, γ becomes 0, δ becomes 1, ϵ becomes -1. Label this new list $A(C)$

      ii. For $A_\Sigma(C)$: $\alpha^n_{n-(1/2)}$ becomes -1 or -2, β becomes 0, γ^a_b becomes $b-a$, δ_d becomes d, $\epsilon^{1,*}$ and ϵ^1 become -1, and each ϵ^2 becomes -2

   Then

$$\Delta_m(C) = \sum_i^\infty A_m(C)_i$$

   and

$$\Delta_\Sigma(C) = \sum_i^\infty A_\Sigma(C)_i$$

If Δ_Σ or Δ_m are well-defined, they are easily computable in the sense that the actions on the propagation of a chain can run forever, from which we can form the lists given in definition 11. The way we should interpret the difference in length is that the digits are moving to the *right* of the chain, while the way we should interpret the difference in size is that for an arbitrary large cut off point, the expected value difference in size should be negative.

Let us examine these difference for infinite chains. Δ_Σ is certainly well-defined in the infinite case as there is no more active end to a chain and hence a chain can either stay the same size or shrink:

Proposition 1: Difference of Size Expected Value: Infinite Chains

Let $C \in \mathcal{C}_\infty$ be a normal chain. Then the expected value of the difference of size is:

$$E(\Delta_\Sigma(C)) = \frac{1}{4} \cdot (-1) + -\frac{2}{3} = -\frac{11}{12}$$

Proof **:**
For $E(\Delta_\Sigma)$, we take advantage of the fact that there isn't the notion of an *active end* anymore, meaning the calculations of cases inactive ends and (5) in theorem 2 are the only ones that need to be considered. The calculations are essentially all in this table:

1 in first position	n \ge 3 in first position		# of 2 in head	chain rep	prob	subtracting	expected value
0.5	0.25		0	~(1/n)	0.75	0	0
			1	~2~(1/n)	0.1875	-2	-0.375
			2	~2~2~(1/n)	0.046875	-4	-0.1875
			3	...	0.0117188	-6	-0.0703125
			4		0.0029297	-8	-0.0234375
			5		0.0007324	-10	-0.007324219
			6		0.0001831	-12	-0.002197266
			7		4.578E-05	-14	-0.000640869
			8		1.144E-05	-16	-0.000183105
			9		2.861E-06	-18	-5.14984E-05
			10		7.153E-07	-20	-1.43051E-05
			11		1.788E-07	-22	-3.93391E-06
			12		4.47E-08	-24	-1.07288E-06
			13		1.118E-08	-26	-2.90573E-07
			14		2.794E-09	-28	-7.82311E-08
			15		6.985E-10	-30	-2.09548E-08
			16		1.746E-10	-32	-5.58794E-09
infinite chain, always acts like inactive end:			17		4.366E-11	-34	-1.4843E-09
E(sum)	E(sum)						
0	-1			should be 1:	1	E(e^2)	-0.666666666

Figure 1.16: expected value given $1/n$ in 1st position, and for # of 2's

By combining results from theorem 2 where we always take the case where the last digit is considered inactive, we get:

$$E(\Delta_\Sigma(C)) - \frac{1}{4} \cdot (-1) + -\frac{2}{3} = -\frac{11}{12}$$

as we sought to show.

Δ_m requires more work, but it is doable. The key idea in computing its expected value is done by translating the distribution of digits to the distribution of ϵ-actions and δ-actions:

Proposition 2: Difference of Length Expected Value: Infinite Chains

Let $C \in \mathcal{C}_\infty$ be a chain constructed by following the probability distribution put on the digits. Then the expected value of the difference of length is:

$$E(\Delta_m(C)) = -\frac{1}{5}$$

Proof **:**
This comes from calculating the probability of an element being consumed for a collapse along with the probability of the new element crated by the termination of a propagation, namely this comes from the calculations of the ϵ's and the δ's. We shall calculate the expected value as we add more digits. By theorem 2, we see that the state of the digits is strongly tied to the blocks, and hence we shall be calculating the expected value as we add more blocks. The calculations

become repetitive, hence the detail of the first steps shall be shown, and then the rest of the steps shall be omitted (these have been verified empirically as well).

Let C be a random infinitely long normal chain. Let us first deal with the head of the chain, which goes until the next digit is $n \geq 3$. As we are working with a normal chain, we have probability 1 that such a digit will be reached. Given the chain start with a 1, then the chain immediately starts in an active state when the 1 is consumed, removed one digit. Then we add a digit as there must be a δ-action by theorem 1 when the propagation encounters an $n \geq 3$, thus the removed 1 is (a.s.) guaranteed to be canceled out by an added digit. If the chain starts with an $n \geq 3$, then the propagation immediately becomes inactive, and no digits are added or removed (we shall deal with how inactive propagation adds digits after analyzing the head). Finally, there is a 25% that the chain starts with a 2. The number of 2's will change the number of digit removed from the chain. Combining all these, we get:

Front of chain rep	diff in length	prob	expected val.
~(1/n)~...	0	0.75	0
~2~(1/n)~...	-1	0.1875	-0.1875
~2~2~(1/n)...	-2	0.046875	-0.09375
...	-3	0.0117188	-0.035156
	-4	0.0029297	-0.011719
	-5	0.0007324	-0.003662
	-6	0.0001831	-0.001099
	-7	4.578E-05	-0.00032
	-8	1.144E-05	-9.16E-05
	-9	2.861E-06	-2.57E-05
	-10	7.153E-07	-7.15E-06
	-11	1.788E-07	-1.97E-06
	-12	4.47E-08	-5.36E-07
	-13	1.118E-08	-1.45E-07
	-14	2.794E-09	-3.91E-08
	-15	6.985E-10	-1.05E-08
	-16	1.746E-10	-2.79E-09
	-17	4.366E-11	-7.42E-10
	-18	1.091E-11	-1.96E-10
	-19	2.728E-12	-5.18E-11
	-20	6.821E-13	-1.36E-11
	-21	1.705E-13	-3.58E-12
	-22	4.263E-14	-9.38E-13
	-23	1.066E-14	-2.45E-13
	-24	2.665E-15	-6.39E-14
	-25	6.661E-16	-1.67E-14
	should be 1 ->	1	-0.333333

Figure 1.17: expected value given front of chain

Next, we must be more precise with the case of 1. As seen in table 1.17, we get a $(1/n)$ block in front of each possible chain of 2. We develop the possible values until we reach the next n. This starts a propagation which removes a digit which will be added back in when the propagation becomes inactive, however there are many possible sizes of blocks in between. For each block of size m, there are multiple ways combination's of $1/2$ before encountering $n \geq 3$. This is a lot of book-keeping, here is a portion of the calculations:

1	~1~n	0.125	0.125	0	0	0	0	0	0	0	0	0	0
2	~1~(1/2)~n	0.09375	0.0625	0.03125	0	0	0	0	0	0	0	0	0
3	~1~(1/2)~(1/2)~n	0.0703125	0.03125	0.03125	0.0078125	0	0	0	0	0	0	0	0
4	...	0.05273438	0.015625	0.0234375	0.01171875	0.00195313	0	0	0	0	0	0	0
5		0.03955078	0.0078125	0.015625	0.01171875	0.00390625	0.00048828	0	0	0	0	0	0
6		0.02966309	0.00390625	0.00976563	0.00976563	0.00488281	0.0012207	0.00012207	0	0	0	0	0
7		0.02224731	0.00195313	0.00585938	0.00732422	0.00488281	0.00183105	0.00036621	3.0518E-05	0	0	0	0
8		0.01668549	0.00097656	0.00341797	0.00512695	0.00427246	0.00213623	0.00064087	0.00010681	7.6294E-06	0	0	0
9		0.01251411	0.00048828	0.00195313	0.00341797	0.00341797	0.00213623	0.00085449	0.00021362	3.0518E-05	1.90735E-06	0	0
10		0.00938559	0.00024414	0.00109863	0.00219727	0.00256348	0.00192261	0.0009613	0.00032043	6.8665E-05	8.58307E-06	4.7684E-07	0
11		0.00703919	0.00012207	0.00061035	0.00137329	0.00183105	0.00160217	0.0009613	0.00040054	0.00011444	2.14577E-05	2.3842E-06	1.1921E-07
12		0.00527939	6.1035E-05	0.00033569	0.00083923	0.00125885	0.00125885	0.0008812	0.0004406	0.00015736	3.93391E-05	6.5565E-06	6.5565E-07
13		0.00395954	3.0518E-05	0.00018311	0.00050354	0.00083923	0.00094414	0.00075531	0.0004406	0.00018883	5.90086E-05	1.3113E-05	1.967E-06
14		0.00296966	1.5259E-05	9.9182E-05	0.00029755	0.0005455	0.00068188	0.00061369	0.00040913	0.00020456	7.67112E-05	2.1309E-05	4.2617E-06
15		0.00222724	7.6294E-06	5.3406E-05	0.00017357	0.00034714	0.00047731	0.00047731	0.00035799	0.00020456	8.94964E-05	2.9832E-05	7.458E-06
16		0.00167043	3.8147E-06	2.861E-05	0.00010014	0.00021696	0.00032544	0.00035799	0.00029832	0.00019178	9.5889E-05	3.729E-05	1.1187E-05
17		0.00125282	1.9073E-06	1.5259E-05	5.722E-05	0.00013351	0.00021696	0.00026035	0.00023866	0.00017047	9.5889E-05	4.2617E-05	1.4916E-05
18		0.00093962	9.5367E-07	8.1062E-06	3.2425E-05	8.1062E-05	0.00014186	0.00018442	0.00018442	0.0001449	9.05618E-05	4.5281E-05	1.8112E-05
19		0.00070471	4.7684E-07	4.2915E-06	1.8239E-05	4.8637E-05	9.1195E-05	0.00012767	0.00013831	0.00011855	8.15056E-05	4.5281E-05	2.0376E-05
20		0.00052854	2.3842E-07	2.265E-06	1.0192E-05	2.8878E-05	5.7757E-05	8.6635E-05	0.00010107	9.3855E-05	7.03912E-05	4.3017E-05	2.1508E-05
21		0.0003964	1.1921E-07	1.1921E-06	5.6624E-06	1.6987E-05	3.6098E-05	5.7757E-05	7.2196E-05	7.2196E-05	5.86594E-05	3.9106E-05	2.1508E-05
22		0.0002973	5.9605E-08	6.2585E-07	3.1292E-06	9.9093E-06	2.2296E-05	3.7903E-05	5.0537E-05	5.4147E-05	4.73787E-05	3.4218E-05	2.0531E-05
23		0.00022298	2.9802E-08	3.2783E-07	1.7211E-06	5.7369E-06	1.3625E-05	2.4525E-05	3.4744E-05	3.9708E-05	3.72261E-05	2.8954E-05	1.882E-05
24		0.00016723	1.4901E-08	1.7136E-07	9.425E-07	3.2987E-06	8.2469E-06	1.5669E-05	2.3504E-05	2.854E-05	2.854E-05	2.3783E-05	1.6648E-05
25		0.00012542	7.4506E-09	8.9407E-08	5.1409E-07	1.885E-06	4.9481E-06	9.8962E-06	1.5669E-05	2.0146E-05	2.1405E-05	1.9027E-05	1.427E-05
26		9.4068E-05	3.7253E-09	4.6566E-08	2.794E-07	1.071E-06	2.9453E-06	6.1851E-06	1.0309E-05	1.399E-05	1.5739E-05	1.4865E-05	1.1892E-05
27		7.0551E-05	1.8626E-09	2.4214E-08	1.5134E-07	6.0536E-07	1.7404E-06	3.8289E-06	6.7006E-06	9.5722E-06	1.1367E-05	1.1367E-05	9.662E-06
28		5.2913E-05	9.3132E-10	1.2573E-08	8.1724E-08	3.4051E-07	1.0215E-06	2.3496E-06	4.3075E-06	6.4613E-06	8.07659E-06	8.5253E-06	7.6728E-06

Figure 1.18: the probability of the number of 2's in different sizes of head blocks

Combining the results, we get:

Front of chain	diff in length	prob	expected val.		Front of chain	diff in length (	prob	exp val of 2	exp val
~n~	0	0.25	0		~1...n~	0	0.5	0	-0.5
~2~n~...	-1	0.0625	-0.0625		~2~1...n~	-1	0.125	-0.125	-0.25
~2~2~n~...	-2	0.015625	-0.03125		~2~2~1...n~	-2	0.03125	-0.0625	-0.09375
...	-3	0.0039063	-0.0117188		...	-3	0.0078125	-0.0234375	-0.03125
	-4	0.0009766	-0.0039063			-4	0.0019531	-0.0078125	-0.009765625
	-5	0.0002441	-0.0012207			-5	0.0004883	-0.0024414	-0.002929688
	-6	6.104E-05	-0.0003662			-6	0.0001221	-0.0007324	-0.000854492
	-7	1.526E-05	-0.0001068			-7	3.052E-05	-0.0002136	-0.000244141
	-8	3.815E-06	-3.052E-05			-8	7.629E-06	-6.104E-05	-6.86646E-05
	-9	9.537E-07	-8.583E-06			-9	1.907E-06	-1.717E-05	-1.90735E-05
	-10	2.384E-07	-2.384E-06			-10	4.768E-07	-4.768E-06	-5.24521E-06
	-11	5.96E-08	-6.557E-07			-11	1.192E-07	-1.311E-06	-1.43051E-06
	-12	1.49E-08	-1.788E-07			-12	2.98E-08	-3.576E-07	-3.8743E-07
	-13	3.725E-09	-4.843E-08			-13	7.451E-09	-9.686E-08	-1.04308E-07
	-14	9.313E-10	-1.304E-08			-14	1.863E-09	-2.608E-08	-2.79397E-08
	-15	2.328E-10	-3.492E-09			-15	4.657E-10	-6.985E-09	-7.45058E-09
	-16	5.821E-11	-9.313E-10			-16	1.164E-10	-1.863E-09	-1.97906E-09
	-17	1.455E-11	-2.474E-10			-17	2.91E-11	-4.948E-10	-5.23869E-10
	-18	3.638E-12	-6.548E-11			-18	7.276E-12	-1.31E-10	-1.38243E-10
	-19	9.095E-13	-1.728E-11			-19	1.819E-12	-3.456E-11	-3.63798E-11
	-20	2.274E-13	-4.547E-12			-20	4.547E-13	-9.095E-12	-9.54969E-12
	-21	5.684E-14	-1.194E-12			-21	1.137E-13	-2.387E-12	-2.50111E-12
	-22	1.421E-14	-3.126E-13			-22	2.842E-14	-6.253E-13	-6.53699E-13
	-23	3.553E-15	-8.171E-14			-23	7.105E-15	-1.634E-13	-1.7053E-13
	-24	8.882E-16	-2.132E-14			-24	1.776E-15	-4.263E-14	-4.44089E-14
	-25	2.22E-16	-5.551E-15			-25	4.441E-16	-1.11E-14	-1.15463E-14
		0.3333333	-0.1111111				0.6666667	-0.2222222	-0.888888889
				total prob	1				
				exp val	-1				

Figure 1.19: Combined Results

Thus:

$$E(\text{from start to next } n) = -1$$

We have now calculated the expected value up to the first n (which might also be at the front if the first digit is an $n \geq 3$). Now, depending on the number of inactive elements in a row, we start adding elements. As 2 are in an inactive state, we shall keep adding digits until we encounter the next 1.

Computing:

P(new digit)	probabiltiy	# new digits	expected val.
~n~1~	0.125	0	0
~n~(2/n)~1~	0.0625	1	0.0625
~n~(2/n)~(2/n)~1~	0.03125	2	0.0625
...	0.015625	3	0.046875
	0.0078125	4	0.03125
	0.0039063	5	0.01953125
	0.0019531	6	0.01171875
	0.0009766	7	0.00683594
	0.0004883	8	0.00390625
	0.0002441	9	0.00219727
	0.0001221	10	0.0012207
	6.104E-05	11	0.00067139
	3.052E-05	12	0.00036621
	1.526E-05	13	0.00019836
	7.629E-06	14	0.00010681
	3.815E-06	15	5.722E-05
	1.907E-06	16	3.0518E-05
	9.537E-07	17	1.6212E-05
	4.768E-07	18	8.5831E-06
	2.384E-07	19	4.53E-06
	1.192E-07	20	2.3842E-06
	5.96E-08	21	1.2517E-06
	2.98E-08	22	6.5565E-07
	1.49E-08	23	3.4273E-07
	7.451E-09	24	1.7881E-07
	3.725E-09	25	9.3132E-08
	1.863E-09	26	4.8429E-08
	9.313E-10	27	2.5146E-08
	4.657E-10	28	1.3039E-08
	2.328E-10	29	6.7521E-09
	1.164E-10	30	3.4925E-09
	5.821E-11	31	1.8044E-09
	2.91E-11	32	9.3132E-10
	1.455E-11	33	4.8021E-10
	7.276E-12	34	2.4738E-10
	3.638E-12	35	1.2733E-10
	1.819E-12	36	6.5484E-11
	9.095E-13	37	3.3651E-11
	4.547E-13	38	1.728E-11
	2.274E-13	39	8.8676E-12
	1.137E-13	40	4.5475E-12
	5.684E-14	41	2.3306E-12
	0.25		0.25

Figure 1.20: Computing Adding Digits Expected Value

Expanding then the computations from table 1.19, we get:

Front of chai	diff in length	prob	expected val.		Front of chain rep	diff in length	prob	exp val of 2	exp val
~n...1~	0	0.25	0.25		~1...n...1~	0	0.5	0	-0.25
~2~n...1~...	-1	0.0625	0		~2~1...n...1~	-1	0.125	-0.125	-0.125
~2~2~n...1~.	-2	0.015625	-0.015625		~2~2~1...n...1~	-2	0.03125	-0.0625	-0.0625
...	-3	0.0039063	-0.007813		...	-3	0.0078125	-0.023438	-0.023438
	-4	0.0009766	-0.00293			-4	0.0019531	-0.007813	-0.007813
	-5	0.0002441	-0.000977			-5	0.0004883	-0.002441	-0.002441
	-6	6.104E-05	-0.000305			-6	0.0001221	-0.000732	-0.000732
	-7	1.526E-05	-9.16E-05			-7	3.052E-05	-0.000214	-0.000214
	-8	3.815E-06	-2.67E-05			-8	7.629E-06	-6.1E-05	-6.1E-05
	-9	9.537E-07	-7.63E-06			-9	1.907E-06	-1.72E-05	-1.72E-05
	-10	2.384E-07	-2.15E-06			-10	4.768E-07	-4.77E-06	-4.77E-06
	-11	5.96E-08	-5.96E-07			-11	1.192E-07	-1.31E-06	-1.31E-06
	-12	1.49E-08	-1.64E-07			-12	2.98E-08	-3.58E-07	-3.58E-07
	-13	3.725E-09	-4.47E-08			-13	7.451E-09	-9.69E-08	-9.69E-08
	-14	9.313E-10	-1.21E-08			-14	1.863E-09	-2.61E-08	-2.61E-08
	-15	2.328E-10	-3.26E-09			-15	4.657E-10	-6.98E-09	-6.98E-09
	-16	5.821E-11	-8.73E-10			-16	1.164E-10	-1.86E-09	-1.86E-09
	-17	1.455E-11	-2.33E-10			-17	2.91E-11	-4.95E-10	-4.95E-10
	-18	3.638E-12	-6.18E-11			-18	7.276E-12	-1.31E-10	-1.31E-10
	-19	9.095E-13	-1.64E-11			-19	1.819E-12	-3.46E-11	-3.46E-11
	-20	2.274E-13	-4.32E-12			-20	4.547E-13	-9.09E-12	-9.09E-12
	-21	5.684E-14	-1.14E-12			-21	1.137E-13	-2.39E-12	-2.39E-12
	-22	1.421E-14	-2.98E-13			-22	2.842E-14	-6.25E-13	-6.25E-13
	-23	3.553E-15	-7.82E-14			-23	7.105E-15	-1.63E-13	-1.63E-13
	-24	8.882E-16	-2.04E-14			-24	1.776E-15	-4.26E-14	-4.26E-14
	-25	2.22E-16	-5.33E-15			-25	4.441E-16	-1.11E-14	-1.11E-14
		0.3333333	0.2222222				0.6666667	-0.222222	-0.472222
				total prob	1				
				exp val	-0.25				

Figure 1.21: Computing Expected Value up to next block

Thus:

$$E(\text{up to next block}) = -0.25$$

We then do the same thing except we shall go on until we get to the next n, and keep updating the expected value of the chains as we add more blocks by repeating the above process. This process shall converge to:

$$E(\Delta_m) = -0.2 = -\frac{1}{5}$$

as we sought to show.

1.4 Arbitrarily Long Finite Chains

Let us now see how we can apply this information to find some information about finite chains. There are two mains ideas: we can either try to show the length must shrink, or that the size must shrink. We try both methods in this section and show what we would need to finish the proof of the Collatz Conjecture. We shall then give some conjectured probability distributions on the digits on the tail motivated by empirical data, and show how proving these conjectured probabilities are enough is enough to show the Collatz conjecture.

Note that when we reference the tail of the chain, we are referencing the average *tail block* (see definition 7).

1.4.1 Length Shrinks

Though Δ_Σ has a strong dependency on the probability of the last digit, the expected value of the difference of length Δ_m has a much weaker dependency on the tail; we only require that the chaos of the tail stays "near" the tail. This section will give my ideas on how to deal with this.

Before proceeding it is worth knowing where the intuition behind why the chaos of the tail is limited to only a few digits. We can program Collatz arithmetic on a chain (if we represented these chains as numbers, the computations quickly exceed integer limits, and `BigInt` computations will take over your RAM), and then keep track of the values of $1/2/n$ of the last ith digits after a certain amount of iterations, we notice the following (explanation bellow image):

the running average value	max	-1 pos		max	-2 pos		max	-3 pos	
of the -nth digit	41.52	26.34	32.26	47.86	25.66	27.2	49.68	25.52	25.92
of 100 chains	min			min			min		
of length 5000	41.5	26.24	32.16	47.44	25.3	26.54	48.88	24.78	25.02
after 1000 iterations	average			average			average		
	41.50405063	26.3032911	32.1926582	47.64734177	25.5144304	26.8382278	49.26303797	25.1810127	25.5559494
	diff (.5/.25/.25)	total:	16.9918987	diff (.5/.25/.25)	total:	4.70531646	diff (.5/.25/.25)	total:	1.47392405
	8.495949367	1.30329114	7.19265823	2.352658228	0.51443038	1.83822785	0.736962025	0.18101266	0.55594937
	41.5	26.3	32.2						
	max	-4 pos		max	-5 pos		max	-6 pos	
	50.92	26.08	26.08	50.86	25.7	26.28	51.58	26	27
	min			min			min		
	48.62	23.96	24.52	48.9	24.26	23.88	48.48	24.2	23.44
	average			average			average		
	49.688	25.117	25.195	49.9306	25.043	25.0264	49.9484	25.0166	25.035
	diff (.5/.25/.25)	total:	0.624	diff (.5/.25/.25)	total:	0.1388	diff (.5/.25/.25)	total:	0.1032
	0.312	0.117	0.195	0.0694	0.043	0.0264	0.0516	0.0166	0.035
	max	-7 pos							
	51.24	25.96	26.14						
	min								
	48.6	24	23.76						
	average								
	50.042	24.9706	24.9874						
	diff (.5/.25/.25)	total:	0.084						
	0.042	0.0294	0.0126						

Figure 1.22: $-i$th digit, valuable statistics

To explain the image, the program kept track of the ith last digit, or -ith digit, the last 7 digits of 100 random chains of length 5000 being iterated 1000 times. In the image, there are 7 boxes each titled $-$`ith pos` for $i = 1, ..., 7$. The box has the maximum, minimum, and average probability of the ith digit being $1, 2, n$. The `diff` row represents how far away the average value is from the usual 50/25/25 probabilities, and the total sums these differences (i.e. it is the variance). The reader can see how stable the probabilities are, as well as how few digits are needed in order for the probabilities to converge. Naturally, this is just a sample space and a relatively small one, but the higher variance comes from larger numbers which have exponentially lower probabilities of appearing, and so this sample space is representative of the general behaviour (though I did not prove this).

Notice that as we get further away from the last digit, the distribution of the digits seem to normalize back to $50\%, 25\%, 25\%$ (by the 7th digit, we are cumulatively off by $< 0.1\%$ from the normal probability values). This is excellent, as that means that theorem 3 should apply almost everywhere for sufficiently large chains.

The main idea behind why the probabilities normalize is that as we move away from the tail, the digits of the chain starts shifting as the length changes to the right, and hence there comes a point where we can consider the effects of the global probability.

The following theorem encapsulates these ideas and my intuition behind why it is theoretically true:

Theorem 4: Distribution a.e. Digit

Let $C \in \mathcal{C}_N$ be an arbitrary non-trivial chain of length $\leq N$ for some arbitrary large N with sequence $(C_0, C_1, ...)$. Consider the functionals $\pi_j(C_i) = n_{i,j}$ which returns the jth digit of the chain C_i. Then as $N \to \infty$:

$$P(\pi_j(C_i) = 1) = 50\% \qquad P(\pi_j(C_i) = 2) = 25\% \qquad P(\pi_j(C_i) \geq 3) = 25\%$$

for all but the last few digits. In other words, given any propagation, on average, the distribution of 1's 2's and n's in a chain is the same as picking an arbitrary chain; F does not alter the probability as it is iterated over.

Incomplete This proof is incomplete (I will point out where).

***Proof* :**

let $C_0 \in \mathcal{C}_N$ be an arbitrary chain where N is very large, and let $C_k = F^{\circ k}(C_0)$. Given that the non-normal behaviour happens at the tail of the chain (as $N \to \infty$, the digits with non-normal behaviour approach 0%), then by theorem 3 and corollary 3, we get the desired normal probabilities for the chain. Hence, it suffices to show that only a certain amount of tail-digits remain non normal as $k \to \infty$.

My idea is to notice that the end of the chain has identical behaviour to the behaviour that precedes a large digit n. This can be demonstrated with an illustrative example. We can use a computer to show that:

$$\underbrace{\sim\ 1 \sim\ 1 \cdots \sim 1}_{50 \text{ times}}$$

goes to $\sim$ (i.e. 1) in 209 iteration of F. Then:

$$\underbrace{\sim\ 1 \sim\ 1 \cdots \sim 1}_{50 \text{ times}} \sim (418 + 3)$$

will have the exact same Collatz sequence when taking the sub-sequence given by taking everything that proceeds the digit $421 = 2 \cdot 209 + 3$, and as it takes 209 iterations for the original sequence to go to $\sim$, we see that the digit 421 cannot be deleted (it can at most become 3 by the time the subsequence disappears).

Hence, the irregularity at tail of the chain gets transformed into a question about to what degree we cannot apply the global probability information to the digits before an $n \geq 3$, that is, how consistent is the behaviour of δ_d/ϵ^e to the right of a digit up to the next n (or the end). The local probability only depends on the digits to the left, and any transformed digit not at the tail has a normal distribution by theorem 3 (from table 1.13).

Phrased more succinctly, given an $I \in \mathbb{N}$, if the distribution of ϵ^e and δ_d to the right of the ith last digit for all $i > I$ is close enough with that of table 1.13, then we can conclude the distributions remains normal almost all the chain. I have not finished thinking about how to do this, however I have some ideas.

Let $\sim (a/i) \sim (1/2/n)\|$ represent the last 2 digits of the chain where a/i represents an active or inactive state;. Consider the following table

Probability	Last Digit	2nd Last Behaviour	Result
$P(\sim i \sim n\|) = \frac{1}{12}$	1	$\sim \underline{m} \sim 1$	$E \sim 1$
$P(\sim a \sim n\|) = \frac{1}{6}$	1	$\sim \underline{m} \sim 1$	$E \sim 1$
$P(\sim i \sim 2\|) = \frac{1}{12}$	1	$\sim \underline{m} \sim 1$	$E \sim 1$
$P(\sim a \sim 2\|) = \frac{1}{6}$	$n \geq 4$	$\sim \underline{E} \sim n$	$E \sim n$
$P(\sim i \sim 1\|) = \frac{1}{6}$	3	$\sim \underline{m} \sim 3$	$E \sim 3$
$P(\sim a \sim 1\|) = \frac{1}{3}$	2	$\sim \underline{E} \sim 1$	$E \sim 2$

where m represents the result of either an $n \mapsto (n - [1/2])$ or a 1 from a δ_1, and E represents that the number at that position is normally distributed (E for expected distribution of the digit). Note that in row 4 and 6 we do indeed get an E digit if we correctly consider the possible digits at this position (with similar calculations as done in theorem 3). The only problem with the E's is that after the 2nd iteration, we do not take into account the new probabilities of $n = 3, 4, 5, ...$ which are no longer uniform; via computer computations we can see that the probability for $n = 3$ and $n = 4$ is off by about a percent from 12.5% and 6.25%, and beyond that remains essentially identical to the normal distribution. After multiple iterations, this shifts the theoretical probability away from the computer-gathered data by around 1%.

Nevertheless, keeping this simplification, we can recursively find the updated probabilities of the digits. We get:

$$9/22 \qquad 6/22 \qquad 7/22.$$

Using these probabilities, we re-compute the distribution of δ_d/ϵ^e, and we shall get that it is close enough so that the last ith digit for $i \geq I$ $(I \geq 7)$ are essentially normal.

If we can either prove that the simplifying assumption does not alter the probability too much to deviate from the answer, or eliminate the simplifying assumption, then this theorem would be proved.

Corollary 4: Expected value of difference of Length for finite Chain

Let $C \in \mathcal{C}$ be an arbitrary chain of finite length. Then as $N \to \infty$, the average size decrease goes to

$$\Delta(C) = -0.2$$

under the assumption that the distribution at the tail has a consistent effect on the length when iterating the Collatz function.

***Proof* :**
Let $C \in \mathcal{C}$ with length N. By theorem 4, for N very large the distribution is a.e. normal. Hence, we can apply proposition 2.

Corollary 5: Uniformization for Finite Chain

Let $C_0 \in \mathcal{C}_N$ be a finite chain of length N, and consider its Collatz sequence $(C_0, C_1, ..., C_n, ...)$. Take the sets $D_{i,j}$ which keep track of how many 1's, 2's and $n \geq 3$'s there are at position j of the chain at the ith position of the Collatz Sequence. If C_i has no position j, skip this position when counting. For example, it is possible that $D_{2,3} = \{1 : 2,\ 2 : 1,\ n : 0\}$. which means that in 3rd position for C_0, C_1, C_2, there were two 1's, one 2, and no $n \geq 3$.

Then as $N \to \infty$, for sufficiently large N, the ratio between these three quantities approach

$$0.5 \qquad 0.25 \qquad 0.25$$

for a.e. digits.

For example, if you consider the chain $(\sim\ 1) \times 10'000$ it will either stay the same size or grow for $10'000$ iterations before eventually reaching a "normal" state.

Proof **:**
First, we can use a computer to show that all chains where $\Sigma(C_0) < 10'000$ and $m(C) < 1000$ will go to 1; we do this so that we don't need to work with the edge case of small chains (in size and length) don't behave well under the probabilistic arguments we made above.

By theorem 4, we have C_0 is normal a.e. and remains so as we iterate F. As $m(C_0) > 1000$, we see all but the last ≈ 7 digits have a normal distribution. Then as it is a.e. Uniformally the same distribution, by probability theory we get that the chain will uniformize as F iterates, as we sought to show.

these results are enough to give the following:

Theorem 5: Reduction to Finite Length

Let C_0 be a finite chain. Then it is enough to show the Collatz Conjecture on chains of length

$$m(C_0) < N$$

for an appropriate N.

Proof **:**
First, brute force the solution for all odd numbers representable by a chain C where $\Sigma(C) < 100'000$ and $m(C) < 100$, where we chose these bounds for better uniformization behaviour that we shall point out in the proof. We can thus without loss of generality assume that one of these is true in our proof.

Let $t \in \mathbb{N}$ be a natural number, $T = f^{\circ\ell}(t)$ be the next odd natural number, C_0 be the chain representation of T, and $C_k = F^{\circ k}(C_0)$. For the sake of contradiction, let's assume that there does not exist a k such that $C_k = \sim$ (recall $\sim$ represents 1), more generally due to our brute-forcing, that there does not exist a k such that $\Sigma(C_k) < 100'000$ and $m(C_k) < 100$. Then there must exists a k_1 where C_{k_1} has uniformized for a.e. digits by corollary 5. Then by corollary 4, the expected difference in length of the chain is negative. But if there does not exist a k such that $m(C_k) < 100$, then by uniformity it must be that the expected difference in length is ≥ 0, a contradiction.

The problem about concluding the result further is that it is possible that $m(C_k) < 100$ but $\Sigma(C_k) > 100'000$ and it stays that way. The reason why I chose $\Sigma(C) > 100'000$ is because this will force a higher density of δ_1's that will increase the size of the chain, which must then necessarily decrease. If the Δ_Σ is negative, we can see how this would give the desired conditions to prove the Collatz Conjecture. In the next section, I give my best guess on how to prove that Δ_Σ is negative, and given that is true then we can conclude the Collatz conjecture from there.

Before moving onto analyzing Δ_Σ, I had one more idea in how to use Δ_m to prove the Collatz conjecture without needing to resort to the sum, however I'm not completely sure of the validity of the last part of this proof; I shall point out explicitly when we get to it:

Theorem 6: Chain Embedding Theorem

Let $C_0 \in \mathcal{C}$ be a finite [normal] chain. Then there exists an infinite normal chain C_∞ and a number N where

1. the beginning of C_∞ is C_0, then followed by the digit N
2. N is large enough so that after a finite amount of iterations, the N will not have been consumed, and it shall be at the front of the chain.

Essentially, all of C_0 will be consumed.

Proof **:**
Let $L = m(C_0)$ and $S = \Sigma(C_0)$.

A normal chain C_∞ with the first digits being equal to C_0 certainly exists as C_0 has finitely many digits, so we can choose the infinitely remaining digits in such a way so that C_∞ is normal (simply choose $1/2/3/.../k/...$ with probability $1/2^k$). Let's say that the a.e. uniformization of C_0 take M steps[a] (we can even say it is at most S steps by using the variance, a proof we haven't covered in this paper but mentioned should be doable with standard probability). Then for the very next digit after the digit of C_0, let us for now just let it be some large number $N \geq 5L + 5M + 3$.

The key idea is that there is no longer an idea of last digit, the positions move around uniformally as expected from an infinite normal chain. Furthermore, by proposition 2 we know the expected difference in length is negative, which as mentioned under definition 11 means that on average the digits are moving from left to right, and the 2's at the beginning are simply disappearing.

After $T = 5L + 5M$ iterations, the expected length of the chain would have diminished by $5(L + M)/5 = L + M$. Then either N is in front or it is not (by its size, it couldn't have been consumed). If it is in front, we are done. If it is not, then as C_M is a.e. uniform and C_{5M+5L} had enough iterations to take advantage of this a.e. uniformity, we would get that this contradicts the expected value of the difference, completing the proof. (This last line of reasoning is where I'm not completely sure about my argument).

[a] it the chain goes to 1 in the uniformization, then naturally we are done

Using Sum Functional Instead The above proof required that the expected value of Δ_m is negative in the infinite case. We can repeat the same proof using the sum-functional as it is also negative.

Theorem 7: [Conjectured] Collatz Conjecture Proof using Length

Let $t \in \mathbb{N}$ and f the Collatz Function:

$$f(t) = \begin{cases} \frac{t}{2} & t \text{ is even} \\ 3t+1 & t \text{ is odd} \end{cases}$$

Then there exists a k such that

$$f^{\circ k}(t) = 1$$

Proof **:**
After picking the next odd t and converting it into a chain C_0, this is a direct consequence of theorem 6 as the Collatz sequence for C_0 is a subsequence of the chain C_∞ and becomes $\sim$ in finitely many steps.

1.4.2 Size Shrinks

By theorem 2, the expected value of the difference of size for finite chins is dependent on the tail of the distribution. While Δ_m only required to know the probabilities near the tail, Δ_Σ requires the exact probability at the end. As alluded to in figure 1.15 right under corollary 2, theorem 3 cannot be extended to the tail, not just in that the probabilities are different, but the probabilities are not stable. Given π_{-1} returns the last digit, for an arbitrary chain C, we have in general that:

$$P(\pi_{-1}(F(C)) = 1) = 1/3 \qquad P(\pi_{-1}(F(C)) = 2) = 1/3 \qquad P(\pi_{-1}(F(C)) \geq 3) = 1/3$$

Which breaks the nice pattern that we've established. Furthermore, we may heuristically see that the probability at the end of the chain fluctuates. Here is a screenshot of the graph of the probability of the last digit of a random chains of length 5000 as we iterate through the Collatz function:

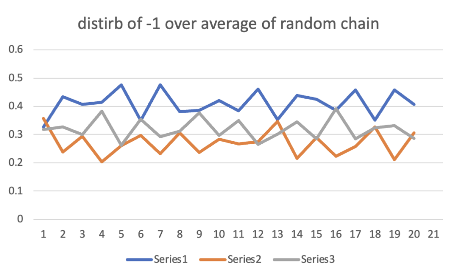

Figure 1.23: Series1 is $P(\pi_{-1} = 1)$, Series2 is $P(\pi_{-1} = 2)$, Series3 is $P(\pi_{-1} \geq 3)$

However, if we take the running average of these probabilities as we iterate the Collatz function, we shall start converging to some stable values, with probabilities that are in fact very simple:

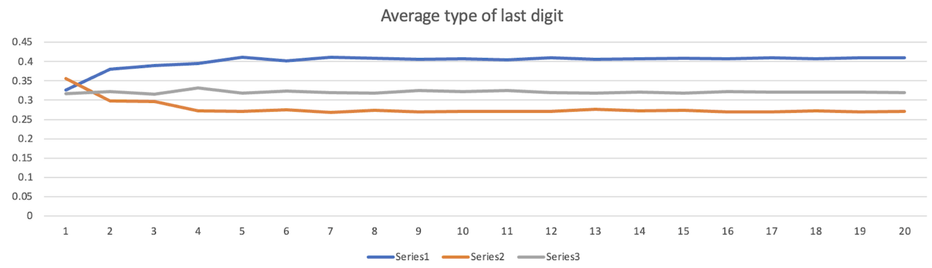

Figure 1.24: Taking the running average from figure 1.23

Part of what makes the calculation of the tail so hard is that the probabilistic computations combine local and global probability, but we cannot combine local probability near the end of the chain as it is always a δ, and the global probability results cannot be applied to the tail as it is no longer an arbitrary position. As of writing this paper there is only a conjecture of what the values ought to be (theorem 8 bellow).

If we can figure out what the eventual probability of an arbitrary chain having an active or inactive component as the last digit, we can invoke theorem 2 to find the eventual average difference between each iteration of the Collatz Function. If this average is negative, by the uniformization results proven earlier it would be enough to conclude the Collatz conjecture. To do this, we need to know how often the last digit is active/inactive. If we look at heuristic data, we will see that the final digit ought to have:

$$P(\text{last digit active}) = 0.6 \qquad P(\text{last digit inactive}) = 0.4$$

If we take this as fact, we can use theorem 2 to get that average decrease in size of a chain ought to be -0.4 (proven in corollary 7). This can again be heuristically verified.

Tail Distribution Ideas

I have done many attempts at finding the probability distribution of the last digit, with the following steps being the closest to getting the desired 0.6/0.4 probability:

1. Given some natural number N, take all tails of length N.
2. Extend these tails (i.e. increase their length) so that after applying the Collatz function, the length remains N.
3. After applying the Collatz Function, compute what is the probability of achieving each tail
4. Update the distribution for the $n \geq 3$ numbers.
5. Calculate the probability of the last digit being active/inactive.
6. Now iterate this process. If all goes well, this process should converge.

So far, I wrote a program that almost does this; it does everything except update the distribution for $n \geq 3$ digits. At the moment, it assumed digits $n \geq 3$ are distributed normally which we can verify heuristically is not the case. With the assumption that n has the same distribution, we converge to (see appendix A)

$$\text{active: } 0.5777777777777777 \qquad \text{inactive: } 0.4222222222222222$$

which is 26/45 and 19/45 respectively. It is off by a little less than 3% for each percentage. In fact, by looking at appendix A, we see that in the 2nd iteration we essentially get 0.6 and 0.4,

$$\text{active: } 0.6058213975694444 \qquad \text{inactive: } 0.3941786024305556$$

It is my guess that not taking into account different sizes of n (i.e. lumping them all together) is causing the sequence to diverge a bit after the 2nd iteration.

If the 0.6/0.4 probabilities can be proven, we can have the following theorem:

Theorem 8: Tail Active/Inactive Distribution

For an arbitrary chain C and Collatz Sequence $(C, F(C), ..., F^{\circ k}(C))$, the probability of the last digit being active or inactive converges to:

$$P(\text{last digit active}) = 0.6 \qquad P(\text{last digit inactive}) = 0.4$$

which converges in less than 100 iterations.

Proof **:**
The best guess at what would be a proof of this result is the program in the Appendix.

An immediate corollary we shall get form this theorem is the following:

Corollary 6: Tail Distribution

For an arbitrary chain C and Collatz Sequence $(C, F(C), ..., F^{\circ k}(C))$, the probability of the tail-digits converges to:

$$\begin{array}{ccc}
P(\pi_{-1} = 1) \approx 41.5\% & P(\pi_{-1} = 2) \approx 26.3\% & P(\pi_{-1} \geq 3) \approx 32.2\% \\
P(\pi_{-2} = 1) \approx 47.6\% & P(\pi_{-2} = 2) \approx 25.5\% & P(\pi_{-2} \geq 3) \approx 26.8\% \\
P(\pi_{-3} = 1) \approx 49.3\% & P(\pi_{-3} = 2) \approx 25.2\% & P(\pi_{-3} \geq 3) \approx 25.5\% \\
P(\pi_{-4} = 1) \approx 49.7\% & P(\pi_{-4} = 2) \approx 25.1\% & P(\pi_{-4} \geq 3) \approx 52.2\% \\
P(\pi_{-5} = 1) \approx 49.9\% & P(\pi_{-5} = 2) \approx 25.04\% & P(\pi_{-5} \geq 3) \approx 25.03\%
\end{array}$$

and the position form 6-10 rapidly converge to 0.5/0.25/0.25.

Proof **:**
This would be the conclusion from the above theorem since we would be calculating the tails probabilities and can keep track of the tails digit distributions. The results shown above are what was found via heuristic methods. What my (incomplete)[a] program found is off by $\sim$ 2-4% per digit.

[a] As mentioned before, it does not take into account the changing probabilistic values of the n's

Corollary 7: Average Size Change

Let C be an arbitrary chain and $(C, F(C), ..., F^{\circ k}(C), ...)$ be the Collatz sequence associated to it. Then eventually, the average size shall change by:

$$-0.4$$

Proof **:**
By using Theorem 2, theorem 4, and theorem 8, the expected value of all chains starting with $1/n$ and ending with a/i could be calculated, in addition to adding all possible $\sim$ 2 in-front. Say each collection of chains that produce the same difference is collected in the set $\mathcal{T}$. Then if $\Delta_T = \sum(F(T)) - \sum(T))$ we must compute:

$$\sum_{T \in \mathcal{T}} P(T) \cdot \Delta_T$$

Delta_T	chain	prob	expected value
	(1/n)-(a/i)	0.75	0.2
-1	-2-1-..-a	0.075	-0.075
-2	-2-1-..-i	0.05	-0.1
-2	-2-n-..-a	0.0375	-0.075
-3	-2-n-..-i	0.025	-0.075
-3	-2-2-1-..-a	0.01875	-0.05625
-4	-2-2-1-..-i	0.0125	-0.05
-4	-2-2-n-..-a	0.009375	-0.0375
-5	-2-2-n-..-i	0.00625	-0.03125
-5	-2-2-2-1-..-a	0.0046875	-0.0234375
-6	-2-2-2-1-..-i	0.003125	-0.01875
-6	-2-2-2-n-..-a	0.00234375	-0.0140625
-7	-2-2-2-n-..-i	0.0015625	-0.0109375
-7	-2x4-1-..-a	0.001171875	-0.008203125
-8	-2x4-1-..-i	0.00078125	-0.00625
-8	-2x4-n-..-a	0.000585938	-0.0046875
-9	-2x4-n-..-i	0.000390625	-0.003515625

Figure 1.25: First few values of the calculation

The formal calculations shall be written down here later, for now it is import to know that after iterating through the first 60 blocks, we have an expected value of -0.399999999985, in other words it is converging to -0.4:

-26		2.23517E-09	-5.81145E-08
-27		1.49012E-09	-4.02331E-08
-27		1.11759E-09	-3.01749E-08
-28		7.45058E-10	-2.08616E-08
-28		5.58794E-10	-1.56462E-08
-29		3.72529E-10	-1.08033E-08
-29		2.79397E-10	-8.10251E-09
-30		1.86265E-10	-5.58794E-09
			-0.399999985

Figure 1.26: after 60 iterations

which is as we sought to show.

Given the assumptions and consequence of what's laid out above, we would be able to conclude the conjecture by the following reasoning:

Theorem 9: [Conjectured] Collatz Conjecture Proof using Size

Let $t \in \mathbb{N}$ and f the Collatz Function:

$$f(t) = \begin{cases} \frac{t}{2} & t \text{ is even} \\ 3t+1 & t \text{ is odd} \end{cases}$$

Then there exists a k such that

$$f^{\circ k}(t) = 1$$

Proof **:**
Let us now assume we have a Collatz sequence for which there does not exist a $K \in \mathbb{N}$ such that $f^{\circ K}(n) = 1$, that is the chain never "reaches" 1[a]. Then the chain must uniformize as it grows for long enough that theorem 4 and corollary 6 to dictate the probabilities of the digits. Then as each chain has a finite sum and never reaches zero, the average difference must be ≥ 0. However, by corollary 7, the average sum of a uniformized chain is -0.4, a contradiction.

[a]Possible behavior may be looping, growing, chaotically moves around

1.5 Conclusion

The Collatz conjecture is a problem that constantly convinces you that there is a next step forward in the fog while never seeming to clear. This paper went as far as examining the behaviour of the Collatz function on infinite normal chains, and shows how if it can be proven that the uniformization a chain applies a.e. (theorem 4) then we can proceed to conclude the Collatz conjecture using the expected value of the difference in length (theorem 7). If we can prove the probabilistic information about the tail (theorem 8) then we can use the expected value of difference in size to prove the conjecture (theorem 9). Using computer's to gather heuristic data, we can at least see that all the probability claims made in theorem 4 and theorem 8 are supported by data. If anyone reads this paper and has any ideas, feel free to reach out to me at nathanael.srawley@mail.utoronto.ca.

A

Tail Distribution Program

There are many functions that go into calculation the tail distributions. Here is the main function that makes references to a lot of helper functions. The docstring of the function describes the desired behaviour:

```python
def BlockedTailDistribution(iterations: int, length: int) -> dict:
  """Running this function will return an approxmimation of the last digit
  probabilities given the number of iterations and the length of the tails
  being considered. The larger they are, the more accurate the result ought
  to be.

  [INITIALIZATION]
  1. Create:
      a) all tails of length N (say, the list is called Tails). Include their
      probabilities
          ex. {[1, 1, 1]: 0.125, [1, 1, 2]: 0.0625, ..., [300, 300, 300]: 0.015625}
      b) Initialize TailDigitProbs with std digit distrib.
          ex : {-1: {1: 0.5, 2: 0.25, 300: 02.5}, ...}
      c) Initialize the LastDigitStateProb:
          ex: {'a': 2/3, 'i': 1/3}
      c) initialize TailProbs
          ex. {(1, 1, 1): 0.125, (1, 1, 2): 0.0625, ...}
  2. Filter all tails in Tails with Active twos into ActiveTwoTails.
  3. Now extend to take into account all possible actions:
      a) what is left over from removing ActiveTwoTails, extended normally
      b) for active two tails, extend far enough so that after applying F, it
         will be of length at least N
  4. Combine the two lists as a dict. Say the dict is called ExtendedTails. It is
     a dict bc we will be keeping track of the probabilities of each ExtendedTails
     ex. {(1, 1, 1, 1): #, (1, 2, 1, 1, 1): #, ..., [300, 300, 300]: #}
  5. Take F(ExtendedTails) and make it a dict, called FofTails (F of
      [Extended]Tails). The value of the dictionary will be the F(extended_tail):
      ex. {(1, 1, 1, 1): [1,1,1,2], (1,2,1,1): [3, 1, 2], ..., (300, 300, 300):
  [298, 1, 299, 1, 299, 1]}

  [ITERATION STEPS]
```

```
    Each iteration step represents the # of F's applied before returning the last
    digit prob.

    6. Update TailDigitProbs using FofTails (this is useful for the return
       value, but not the calculations). At the same time, update lastDigitStatus.
    7. Update Tails (be sure to tally appropriately). Using this, update
    ExtendedTails (this is key for the
       calculations of the next iteration)

    [FINAL]
    8. return the LastDigitStateProb dictionary.

    Note that currently, the progrma does not return the TailDigitProbs, however it
    keeps track of it bc I may want to look at these values in the future or
    while debugging

    Args:
        iterations: As space and time are finite, decide how many times the
        progrma iterates.
        length: As memory is finite, calculate the probility for the last
        "length" digitis. The rest will be assumed to always have
        0.5/0.25/0.25. This assumption is valied for large enough "length"

    Returns:
        The probability of a/i (active/inactive) of the last digits.
    """
    # Initialize the data structures keeping track of the data.

    tailDigitProbs: dict[int, dict[str | int, Fraction]] = {}
    for pos in range(1, length + 1):
        tailDigitProbs[-pos] = {
            1: Fraction(1, 2),
            2: Fraction(1, 4),
            "n": Fraction(1, 4),
        }
    lastAIDistrib: dict[str, Fraction] = {'a': Fraction(2,3), 'i': Fraction(1,3)}

    # Only once will tails probability be given by the specific digits. The rest
    # of the time it will be given by tailProb calculations.
    tails = initializeTailProbs(createTails(length))

    # print(fOfExtendedTails)

    # loop ITERATION time:
    for i in range(iterations):
        print(f'ITERATION # : {i+1}')
        # calculate probability of each exented tail using tails
        # NOTE: a current inefficiency: The extTails are being re-calculated,
        # when only the percentages need to be updated. This will be fixed later
        # when needed, rn it just slows down the program for large inputs.
        extendedTails = initializeExtendedTails(tails, tailDigitProbs, length)

        # save those probabilitie in fOfTails
        fOfExtendedTails = nextTails(extendedTails)
        #             extTail          Prob            F(extTail)
        # format: (1, 1, 1, 1): (Fraction(1, 16), (1, 1, 1, 2)),

        # calculate prob of each position
        tailDigitProbs = updateDigitPositionProb(fOfExtendedTails, length)

```

```
        # TEST: debugging
        for pos, value in tailDigitProbs.items():
            print('current position: ' + str(pos))
            totalProb = Fraction(0,1)
            for theType, prob in value.items():
                totalProb += prob
                print(str(theType) + ' : ' + str(float(prob)))
            print(f"totalProb: {totalProb}")
            print()

        # making an excel friendly printout
        # value = tailDigitProbs[-1]
        # print(f"{float(value[1])}, {float(value[2])}, {float(value['n'])}")


        # calc the prog of each tail
        tails = nextTailProb(fOfExtendedTails, length)
        lastAIDistrib = updatelastAIDistrib(fOfExtendedTails)
        print('last distrib:' +  str(float(lastAIDistrib['a'])) + ' , ' + str(float(lastAIDistrib['i'])))

        print('-------------------------')

        # TEST: debugging
        # for tail, probs in tails.items():
        #     print(f"{tail}, prob: {probs}")
        # print()


    # return digit probabilities
    return tailDigitProbs
```

Here is the output of :

`BlockedTailDistribution(20, 7) # 20 iterations, last 7 digits`

```
ITERATION # : 1
current position: -1
1 : 0.3333333333333333
2 : 0.3333333333333333
n : 0.3333333333333333
totalProb: 1

current position: -2
1 : 0.5694444444444444
2 : 0.2152777777777778
n : 0.2152777777777778
totalProb: 1

current position: -3
1 : 0.5185185185185185
2 : 0.24074074074074073
n : 0.24074074074074073
totalProb: 1

current position: -4
1 : 0.558641975308642
2 : 0.22067901234567902
n : 0.22067901234567902
totalProb: 1
```

```

current position: -5
1 : 0.5282921810699589
2 : 0.23585390946502058
n : 0.23585390946502058
totalProb: 1

current position: -6
1 : 0.5517832647462277
2 : 0.22410836762688616
n : 0.22410836762688616
totalProb: 1

current position: -7
1 : 0.5338363054412437
2 : 0.23308184727937814
n : 0.23308184727937814
totalProb: 1

last distrib:0.5208333333333334 , 0.4791666666666667
-------------------------
ITERATION # : 2
current position: -1
1 : 0.4391547309027778
2 : 0.2433810763888889
n : 0.3174641927083333
totalProb: 1

current position: -2
1 : 0.6100452564380787
2 : 0.20376643428096064
n : 0.18618830928096064
totalProb: 1

current position: -3
1 : 0.5239951051311729
2 : 0.26242555217978397
n : 0.21357934268904322
totalProb: 1

current position: -4
1 : 0.5882962701742541
2 : 0.215839024924447
n : 0.19586470490129887
totalProb: 1

current position: -5
1 : 0.545651827002422
2 : 0.24055811668783222
n : 0.2137900563097458
totalProb: 1

current position: -6
1 : 0.5729340255451246
2 : 0.22064923433391917
n : 0.2064167401209562
totalProb: 1

current position: -7
1 : 0.5455822565991274
```

```
2 : 0.2367871212824836
n : 0.21763062211838896
totalProb: 1

last distrib:0.6058213975694444 , 0.3941786024305556
-------------------------
ITERATION # : 3
current position: -1
1 : 0.3671488192836934
2 : 0.32211291152263377
n : 0.31073826919367287
totalProb: 1

current position: -2
1 : 0.6067816128619578
2 : 0.20152026308580354
n : 0.19169812405223874
totalProb: 1

current position: -3
1 : 0.5033766313617394
2 : 0.26194154491258853
n : 0.23468182372567203
totalProb: 1

current position: -4
1 : 0.5959863043518677
2 : 0.21227204903160776
n : 0.19174164661652449
totalProb: 1

current position: -5
1 : 0.5474744539700139
2 : 0.24091830543001413
n : 0.21160724059997194
totalProb: 1

current position: -6
1 : 0.5694298007979722
2 : 0.22128428297780756
n : 0.20928591622422021
totalProb: 1

current position: -7
1 : 0.5453421379395408
2 : 0.23569230665528615
n : 0.21896555540517307
totalProb: 1

last distrib:0.5906957304526749 , 0.40930426954732513
-------------------------
ITERATION # : 4
current position: -1
1 : 0.37764388707112545
2 : 0.2559975639571596
n : 0.3663585489717149
totalProb: 1

current position: -2
1 : 0.5914134394546104
```

```
2 : 0.20801421934521575
n : 0.2005723412001738
totalProb: 1

current position: -3
1 : 0.49051487216528067
2 : 0.2723889575062913
n : 0.23709617032842803
totalProb: 1

current position: -4
1 : 0.5978900909399921
2 : 0.21114353958763074
n : 0.1909663694723771
totalProb: 1

current position: -5
1 : 0.542673468461553
2 : 0.24592568302269688
n : 0.2114008485157502
totalProb: 1

current position: -6
1 : 0.5661950860049343
2 : 0.2221442801167614
n : 0.21166063387830433
totalProb: 1

current position: -7
1 : 0.5440382131308459
2 : 0.23579598620117617
n : 0.22016580066797795
totalProb: 1

last distrib:0.5402145497756403 , 0.4597854502243597
-------------------------
ITERATION # : 5
current position: -1
1 : 0.4301568325870183
2 : 0.270672234505298
n : 0.2991709329076837
totalProb: 1

current position: -2
1 : 0.5940244528095208
2 : 0.2056204842932277
n : 0.20035506289725152
totalProb: 1

current position: -3
1 : 0.46776859506259894
2 : 0.2823718743166058
n : 0.24985953062079527
totalProb: 1

current position: -4
1 : 0.5976267951000072
2 : 0.20822019444546103
n : 0.1941530104545318
totalProb: 1
```

```

current position: -5
1 : 0.5426985463265667
2 : 0.2418997327264257
n : 0.21540172094700766
totalProb: 1

current position: -6
1 : 0.5680214317313089
2 : 0.2217665768035711
n : 0.21021199146511999
totalProb: 1

current position: -7
1 : 0.5448828717657515
2 : 0.23556201547027106
n : 0.21955511276397752
totalProb: 1

last distrib:0.6105736744253858 , 0.3894263255746142
-------------------------
ITERATION # : 6
current position: -1
1 : 0.3624661029538874
2 : 0.30261167819028467
n : 0.3349222188558279
totalProb: 1

current position: -2
1 : 0.5940097321338363
2 : 0.20847889139547884
n : 0.19751137647068479
totalProb: 1

current position: -3
1 : 0.4868215742045899
2 : 0.2635774306345537
n : 0.2496009951608564
totalProb: 1

current position: -4
1 : 0.5962346452425306
2 : 0.21067223512913397
n : 0.19309311962833547
totalProb: 1

current position: -5
1 : 0.5446479443911582
2 : 0.2411513435919012
n : 0.21420071201694066
totalProb: 1

current position: -6
1 : 0.5668020302647171
2 : 0.2226692527243512
n : 0.21052871701093173
totalProb: 1

current position: -7
1 : 0.5439480413763997
```

```
2 : 0.23632877168798047
n : 0.21972318693561976
totalProb: 1

last distrib:0.5586323680383025 , 0.4413676319616975
-------------------------
ITERATION # : 7
current position: -1
1 : 0.40853096296824837
2 : 0.2580272828589134
n : 0.3334417541728382
totalProb: 1

current position: -2
1 : 0.5977355635189312
2 : 0.20526667981759472
n : 0.1969977566634741
totalProb: 1

current position: -3
1 : 0.48318407521060747
2 : 0.27622498497027764
n : 0.24059093981911486
totalProb: 1

current position: -4
1 : 0.5983744540087899
2 : 0.20770558330389569
n : 0.19391996268731448
totalProb: 1

current position: -5
1 : 0.5441122699122931
2 : 0.24269140144246606
n : 0.21319632864524082
totalProb: 1

current position: -6
1 : 0.5678931163564992
2 : 0.22136480306311293
n : 0.21074208058038787
totalProb: 1

current position: -7
1 : 0.543423480903079
2 : 0.23620307897607265
n : 0.2203734401208484
totalProb: 1

last distrib:0.5793998594007885 , 0.42060014059921147
-------------------------
ITERATION # : 8
current position: -1
1 : 0.3934938346024613
2 : 0.2916210612554627
n : 0.314885104142076
totalProb: 1

current position: -2
1 : 0.5948070914596424
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```
2 : 0.20787491960878027
n : 0.19731798893157737
totalProb: 1

current position: -3
1 : 0.4819972176594717
2 : 0.27205630336563935
n : 0.24594647897488897
totalProb: 1

current position: -4
1 : 0.5977488086201077
2 : 0.20854339253756604
n : 0.19370779884232625
totalProb: 1

current position: -5
1 : 0.5448183197030112
2 : 0.24155526172995148
n : 0.21362641856703732
totalProb: 1

current position: -6
1 : 0.5674266658713385
2 : 0.2223120085507094
n : 0.21026132557795216
totalProb: 1

current position: -7
1 : 0.5441725240424239
2 : 0.23608105335537044
n : 0.21974642260220564
totalProb: 1

last distrib:0.5876181109640428 , 0.4123818890359571
-------------------------
ITERATION # : 9
current position: -1
1 : 0.38208539222409715
2 : 0.27659015713264323
n : 0.34132445064325956
totalProb: 1

current position: -2
1 : 0.5940528308486999
2 : 0.20865657285135367
n : 0.19729059629994652
totalProb: 1

current position: -3
1 : 0.4878447001554972
2 : 0.2695999841279894
n : 0.24255531571651343
totalProb: 1

current position: -4
1 : 0.5976043313193933
2 : 0.20918986773802387
n : 0.1932058009425829
totalProb: 1
```

```

current position: -5
1 : 0.5438017457209309
2 : 0.2430149338994269
n : 0.2131833203796422
totalProb: 1

current position: -6
1 : 0.56691677808222
2 : 0.22206618257532593
n : 0.21101703934245414
totalProb: 1

current position: -7
1 : 0.5436631743912321
2 : 0.23610191420847493
n : 0.22023491140029294
totalProb: 1

last distrib:0.5615916845267699 , 0.43840831547323006

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ITERATION # : 19
current position: -1
1 : 0.39092351098931305
2 : 0.27997387265276974
n : 0.3291026163579172
totalProb: 1

current position: -2
1 : 0.5947803390157832
2 : 0.20786511854830228
n : 0.1973545424359145
totalProb: 1

current position: -3
1 : 0.48380450170141454
2 : 0.27186780094173185
n : 0.24432769735685358
totalProb: 1

current position: -4
1 : 0.5979242674029835
2 : 0.20872785750997797
n : 0.19334787508703846
totalProb: 1

current position: -5
1 : 0.5442932584327005
2 : 0.24230097739776055
n : 0.21340576416953894
totalProb: 1

current position: -6
```

```
1 : 0.5672157954950873
2 : 0.22207355517373892
n : 0.21071064933117375
totalProb: 1

current position: -7
1 : 0.5438753421051394
2 : 0.23609227184437545
n : 0.22003238605048508
totalProb: 1

last distrib:0.5748949303809437 , 0.4251050696190563
-------------------------
ITERATION # : 20
current position: -1
1 : 0.3951890991220008
2 : 0.2772112736019135
n : 0.3275996272760857
totalProb: 1

current position: -2
1 : 0.5950170564752371
2 : 0.20760602064004824
n : 0.19737692288471465
totalProb: 1

current position: -3
1 : 0.482928537015872
2 : 0.2731462469005154
n : 0.2439252160836126
totalProb: 1

current position: -4
1 : 0.5980513189580616
2 : 0.20854472221729675
n : 0.1934039588246416
totalProb: 1

current position: -5
1 : 0.5442369821565721
2 : 0.24234493429713316
n : 0.21341808354629468
totalProb: 1

current position: -6
1 : 0.5672979126462777
2 : 0.2219885219483897
n : 0.2107135654053326
totalProb: 1

current position: -7
1 : 0.5438665927771179
2 : 0.2360768470488807
n : 0.22005656017400133
totalProb: 1

last distrib:0.5777240578216297 , 0.4222759421783703
-------------------------
```

Index

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